



Small Unmanned Aircraft System (sUAS) Categorization Framework V2.0

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Abstract

The unmanned aircraft system (UAS) traffic management (UTM) concept is envisioned to enable civilian low-altitude airspace UAS operations by providing services such as airspace design, corridors, dynamic geo-fencing, severe weather and wind avoidance, congestion management, terrain avoidance, route planning and re-routing, separation management, sequencing and spacing, and contingency management.

Categorization of small unmanned aircraft systems (sUASs) seeks to recognize, differentiate, and understand various sUAS from the perspectives of their design, operational, and safety characteristics. These characteristics include, but are not limited to, airframe configurations, aerodynamic characteristics, weight, performance (including speed), automatic flight control logic, degree of flight control autonomy possible, auto-recovery mechanisms, and navigation and guidance capabilities. Within the context of UTM, an sUAS categorization framework has been established on the foundation of understanding the effects of sUAS design and operational characteristics on the trajectory of the sUAS itself and interactions the sUAS may have with other aerial vehicles, people, and structures on the ground. Such a categorization framework allows for sUAS operations to be managed efficiently, considering operating conditions and efficiency and safety objectives, based on distinctive sUAS characteristics recognized according to the established framework.

In this document, categorization methods applied to manned aircraft and existing UAS categorization methods (including model aircraft) were examined first to provide a starting point for additional analysis specific to sUAS characteristics. The sUAS categorization framework was then presented, along with necessary analysis and justification. In contrast to prior UAS categorization studies that have been mostly targeted towards airworthiness certification, resulting in a single categorization in each case, the proposed categorization framework emphasizes its implications on sUAS trajectories. The framework thus includes aircraft configuration, type of flight, flight rules, performance-based navigation, flight and operations control, and flight performance. The categorization framework is then iteratively improved and refined as the understanding of sUAS-specific operational characteristics deepens, and as the trajectory modeling research progresses. Numeric analysis required to support the determination of various categorization criteria are identified for next step studies where possible. Additional studies, including simulation and potential flight test are also suggested for the final determination and incorporation of the categorization framework into UTM operations.

Key Words

Unmanned Aircraft System, Small Unmanned Aircraft System, Traffic Management, Trajectory Prediction, System Safety, Environmental Conditions, Modeling and Simulation

Revision History

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Background

GE Global Research has been tasked by the NASA under contract NNA16BE13C (executed on September 30, 2016) to perform the research entitled “Small Unmanned Aircraft System (sUAS) Trajectory Modeling.”

The objective of this research is to develop and validate agile, scalable, and affordable sUAS trajectory modeling and prediction approaches. The trajectory modeling and prediction approaches are required to be able to support trajectory patterns for diverse airframe configurations, including horizontal take-off and landing (HTOL), vertical flight, and hybrid vertical take-off and landing (VTOL) aircraft [50], under nominal conditions and under a variety of realistic potential hazards, including adverse environmental conditions, and vehicle and system failures.

As part of this effort, the research team is required to develop a categorization framework for sUASs. The categorization is done based on sUAS airframe configurations, aerodynamic characteristics, weight, performance (including speed), automatic flight control logic, degree of flight control autonomy possible, auto-recovery mechanisms, navigation and guidance capabilities, and other factors mutually agreed with NASA. The focus of the classification framework is given to the support of sUAS traffic services. The criteria and rationale for the categorization method are carefully analyzed.

This document provides the outcome of the categorization development effort.

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1 Introduction

For civilian operations in the United States, the term *small unmanned aircraft system* (sUAS) is defined as a system consisting of an unmanned aircraft or unmanned aerial vehicle (UAV) weighing less than 55 lb, and equipment necessary for the safe and efficient operation of that aircraft [18], [27]. Unmanned aircraft of this size have been flown as model aircraft for hobby, recreational, and sports use for many decades. The advancement, increased access by the public, and the reduced cost and size of technologies such as digital communications, digital controls, navigation, and autonomy have enabled fast growing capabilities of sUASs, including model aircraft, that were not imaginable decades ago. However, wide-spread use of sUASs for non-recreational purposes has been limited, particularly for applications requiring operations beyond the visual line of sight (BVLOS) of the sUAS operator.

Driven by the potential societal and economic benefits that can be generated from application of sUASs, the Federal Aviation Administration (FAA) recently published a rule [27] that added Part 107 to Title 14 of the Code of Federal Regulations (14 CFR) [24]. The rule allows routine use of sUASs in the National Airspace System (NAS). Part 107 operations are limited to open airspace at or below 400 ft above ground level (AGL) or within 400 ft from fixed structures, to avoid interference with manned aircraft operations. Additionally, sUAS operations under this rule are restricted to the visual line-of-sight (VLOS) of the remote pilot in command (PIC) or a visual observer in effective communications with the PIC. Although the VLOS restriction may be waived under special conditions, routine BVLOS operations are expected in the future to fully realize the sUAS benefits. In its simplest form, the sUAS operational space can be summarized as shown in Figure 1. This figure illustrates the extent of the sUAS operational space—from the long history of model aircraft VLOS hobby and recreational flying to currently authorized VLOS commercial operations, and to future BVLOS commercial operations. In this figure, the green areas represent currently unregulated (model aircraft) and authorized routine operations (non-recreational VLOS). The light blue areas represent future expansion into routine BVLOS operations yet to be fully tested.

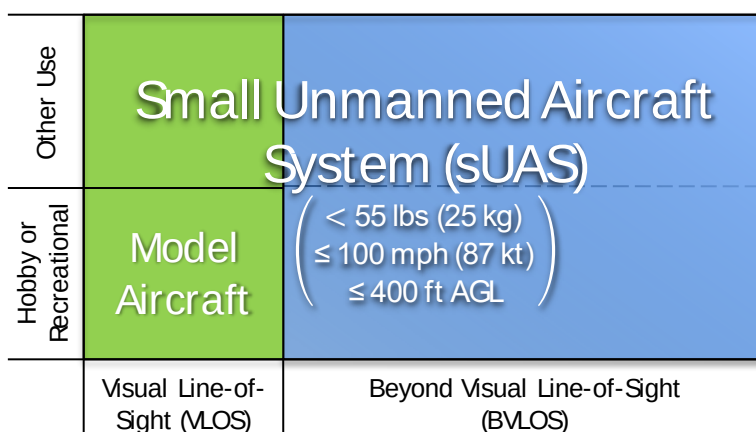


Figure 1: Small UAS (sUAS) operational space defined.

The unmanned aircraft system (UAS) traffic management (UTM) concept is envisioned by the NASA to enable civilian low-altitude airspace and UAS operations, particularly sUAS BVLOS operations, by providing services such as airspace design, corridors, dynamic geofencing, severe

weather and wind avoidance, congestion management, terrain avoidance, route planning and re-routing, separation management, sequencing and spacing, and contingency management [61]. Trajectory modeling and prediction methods are foundational capabilities in support of UTM to achieve its goals. Several challenges need to be addressed in developing such capabilities.

Small UAS design has been evolving. Over time, vehicles with increasingly diverse design, size, flight envelope, and flight characteristics are expected. Vehicles with different airframe configurations have been seen in various sUAS applications. Some examples are 1) horizontal take-off and landing (HTOL) fixed wing aircraft; 2) vertical flight or powered-lift aircraft; and 3) hybrid configurations with combined HTOL and vertical flight characteristics, such as vertical take-off and landing (VTOL) aircraft. With this, it is important to identify how diverse types of sUAS configurations could be modeled and how their trajectories could be planned by the operators and represented UTM.

Also, to support many of the envisioned UTM services and ensure safety of operations, it is important to understand abnormal trajectories that could be generated under a variety of system failure hazards and environmental conditions, as shown by the stack of factors influencing the sUAS trajectory in [Figure 2](#).

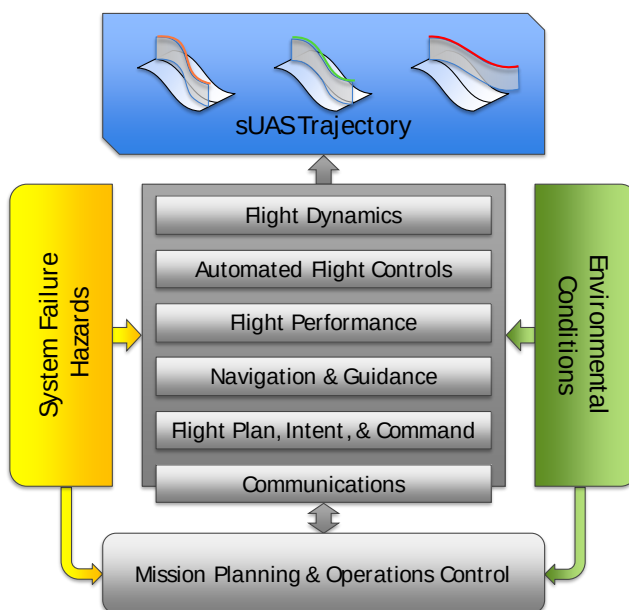


Figure 2: Stack of factors influencing sUAS trajectory.

Categorization of sUASs seeks to recognize, differentiate, and understand various sUASs from the perspectives of their design, operational, and safety characteristics. These characteristics include, but are not limited to, airframe configurations, aerodynamic characteristics, weight, performance (including speed), automatic flight control logic, degree of flight control autonomy possible, auto-recovery mechanisms, and navigation and guidance capabilities. Within the context of UTM and trajectory modeling, an sUAS categorization framework has been established on the foundation of understanding the effects of sUAS design and operational characteristics on the trajectory of the sUAS itself and the interactions the sUAS may have with other aerial vehicles, people, and structures on the ground. Such a categorization framework allows for sUAS operations to be managed efficiently, considering operating conditions and efficiency and safety objectives, based on distinctive sUAS characteristics recognized according to the established framework.

Categorization methods applied to manned aircraft and existing UAS categorization methods (including model aircraft) are examined first to provide a starting point for additional analysis of sUAS-specific characteristics. The categorization framework is expected to be iteratively improved as the understanding of sUAS-specific operational characteristics deepens, and as the trajectory modeling research progresses.

The main outcome of this effort, the proposed sUAS categorization framework was previously published as a conference paper [73] based on Version 1.0 of this document. Due to space limits, that paper only provided a brief high-level summary of existing manned and unmanned aircraft categorization methods, which is presented in full detail in this document. Additionally, the current version of this document is updated with the latest developments in the existing manned and unmanned aircraft categorization methods, particularly the application of performance-based and risk-based standards. The proposed sUAS categorization framework has also been updated to include a new equipage and capability component.

In this document, when a measurement from a foreign or international reference is cited, the unit in the original reference document is quoted directly, with the U.S. equivalent given in parentheses, where applicable. For the proposed new sUAS categorization (Section 3), the U.S. units are used as the primary units.

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2 Existing Manned and Unmanned Aircraft Categorization Methods

This section provides an examination of existing categorization methods applied to manned and unmanned aircraft. Attention is given to design, operations, and safety aspects, and to the implications of these categorization methods. Such an examination is expected to provide a starting point and insights for additional analysis of sUAS-specific characteristics.

2.1 Manned Aircraft Categorization

Manned aircraft have dominated aviation until recently. Comprehensive aircraft categorization methods are in place to recognize, differentiate, and understand various aircraft from the perspectives of their design, operational, and safety characteristics. This section focuses on categorization adopted by the FAA and International Civil Aviation Organization (ICAO), which are believed to be representative of the state of the art.

2.1.1 Aircraft Configuration and Power Plant

Aircraft configuration is the most obvious categorization method. The FAA's definition of aircraft configuration classification (14 CFR Part 1 [24]) are summarized in [Table 1](#). Although not necessarily an exhaustive categorization of all possible aircraft configurations, this summary table does provide a broad grouping of aircraft having similar characteristics in terms of propulsion, flight, or takeoff and landing. In general, each of the configurations listed can be further divided into subcategories. For example, an airplane could be further categorized as conventional wing/tail configuration, canard configuration, and tandem wing configuration. Aircraft performance and dynamics do vary among these subcategories. However, it is also true that aircraft performance and dynamics may vary even more significantly due to differences in design parameters, even for the same sub-configuration. For this reason, [Table 1](#) can be viewed as a good starting point for aircraft configuration-based categorization.

ICAO Doc 8643 [56] defines an aircraft type description scheme to support air traffic service (ATS), particularly to support ATS automation. In the ICAO 8643 definition, an aircraft type is described by the type of the aircraft, the number of engines, and the type of engines, as defined in [Table 2](#), [Table 3](#), [Table 4](#), and [Table 5](#).

[Table 2](#) shows ICAO aircraft type categorization and FAA aircraft class [44]. Each category is assigned a one-letter type code. As shown, the ICAO does not differentiate between a land-based conventional take-off and landing airplane, and land-based thrust deflection jet VTOL airplane.

Normally, an aircraft type designator is assigned to a unique aircraft type, excluding alternate or subtypes when there is no significant difference in performance from the original aircraft type. Six special aircraft types, namely balloons, airships, gliders, micro-/ultra-light aircraft, micro-/ultra-light helicopter, and micro-/ultra-light gyrocopter are also defined by the ICAO to represent aircraft types that have not been assigned an aircraft type designator per ICAO rules. These special aircraft types are listed in [Table 3](#). For these special aircraft types, an aircraft type code is not assigned. The FAA has defined several additional special aircraft types for unmanned and optionally piloted aircraft [43] that were later removed [44], also listed in [Table 3](#).

Table 1: FAA Defined Aircraft Configuration Classification.

Classification		Definition	
Lighter than Air	Balloon	Not engine driven, and that sustains flight through buoyancy.	
	Airship	Engine-driven lighter-than-air aircraft that can be steered.	
Heavier than Air	Kite		Intended to be flown at the end of a rope or cable, and having as its only support the force of the wind moving past its surfaces.
	Fixed Wing	Glider	Flight not depending principally on an engine.
		Airplane	Engine-driven.
	Weight-Shift Control		Powered aircraft with a framed pivoting wing and a fuselage controllable only in pitch and roll by the pilot’s ability to change the aircraft’s center of gravity with respect to the wing.
	Powered Parachute		Comprising a flexible or semi-rigid wing connected to a fuselage so that the wing is not in position for flight until the aircraft is in motion.
	Rotorcraft	Gyroplane	Rotor not engine-driven, except for initial starting, but made to rotate by action of the air when the rotorcraft is moving; and whose means of propulsion, is independent of the rotor system.
		Gyrodyne	Rotors normally engine-driven for takeoff, hovering, and landing, and for forward flight through part of its speed range, and whose means of propulsion, is independent of the rotor system.
		Helicopter	Rotorcraft that, for its horizontal motion, depends principally on its engine-driven rotors for both lift and propulsion.
	Powered-lift		Capable of vertical takeoff, vertical landing, and low speed flight that depends principally on engine-driven lift devices or engine thrust for lift; and on nonrotating airfoil(s) for lift during horizontal flight.

Table 2: ICAO Aircraft Types and FAA Aircraft Classes.

ICAO Type Code	ICAO Category	FAA Class	Explanation
L	Landplane	Fixed-wing	Including temporarily converted floatplane and jet VTOL airplane.
S	Seaplane	\$Fixed-wing	
A	Amphibian	@Fixed0time	
H	Helicopter	Helicopter	
G	Gyrocopter	Gyrocopter	
T	Tilt-wing	Tilt-rotor	Including tilt-rotor aircraft, such as V-22.
		Remotely Piloted	

Table 3: ICAO and FAA Special Aircraft Types.

Type Code	Category	Description
-	Balloon	ICAO designator: BALL
	Airship	ICAO designator: SHIP
	Glider	ICAO designator: GLID
	Micro-/ultra-light Aircraft	ICAO designator: ULAC. Maximum certified take-off mass ≤ 454 kg/1,000 lb; stall speed ≤ 35 kt.
	Micro-/ultra-light Helicopter	ICAO designator: UHEL. Maximum certified take-off mass ≤ 454 kg/1,000 lb
	Micro-/ultra-light Gyrocopter	ICAO designator: GYRO. Maximum certified take-off mass ≤ 454 kg/1,000 lb
	Powered parachute/paraplane.	FAA designator: PARA
	Micro-/ultra-light Optionally Piloted Aircraft	FAA designator (removed [44]): UOPL. Maximum certified take-off weight ≤ 330 lb; stall speed ≤ 35 kt.
	Micro-/ultra-light Optionally Piloted Helicopter	FAA designator (removed [44]): UOPH. Maximum certified take-off weight ≤ 330 lb
	Micro-/ultra-light Remotely Piloted Aircraft	FAA designator (removed [44]): URPL. Maximum certified take-off weight ≤ 330 lb; stall speed ≤ 35 kt.
	Micro-/ultra-light Remotely Piloted Helicopter	FAA designator (removed [44]): URP. Maximum certified take-off weight ≤ 330 lb
	Micro-/ultra-light Autonomous Unmanned Aircraft	FAA designator (removed [44]): UAUL. Maximum certified take-off weight ≤ 330 lb; stall speed ≤ 35 kt.
	Micro-/ultra-light Autonomous Unmanned Helicopter	FAA designator (removed [44]): UAULH. URP. Maximum certified take-off weight ≤ 330 lb

Table 4 shows ICAO number-of-engines categorization. The FAA, however, does not restrict the number to be used to describe the number of engines installed on an aircraft [44]. Table 5 shows ICAO and FAA aircraft engine type categorization. Again, a one-letter class code is used to identify each class or category. Note that the FAA includes Type Code R, Rocket as an engine type.

Table 4: ICAO Number of Engines.

Class Code	Category	Explanation
1, 2, 3, 4, 6, or 8	One-, two-, three-, four-, six-, or eight-engine	FAA does not restrict the numbers to be used.
C	Co-axis Two-engine	For fixed-wing aircraft with two-engines coupled to drive a single propeller system

Table 5: ICAO Aircraft Engine Type Categorization.

Type Code	Category	Explanation
P	Piston Engine	
T	Turboprop/Turboshaft Engine	
J	Jet Engine	
E	Electric Engine	
R	Rocket	FAA only.

With the aircraft type categorization, number of engines, and aircraft engine type categorization defined, a combined three-letter code is used by the ICAO to identify the categorization of an aircraft configuration. For example, the Boeing 787-8 airliner, the Bell 406 helicopter, and the Bell-Boeing V-22 Osprey tilt-rotor aircraft are categorized as L2J, H1T, and T2T, respectively.

2.1.2 Aircraft Categorization in Certification Standards

From the regulation point of view, the most important aircraft categorization is by certification standards. Such a categorization is a grouping of aircraft-based upon intended use or operating limitations. The FAA’s aircraft certification includes a type certificate (TC), which is a design approval issued when it is demonstrated that the aircraft model complies with the applicable regulations (14 CFR Subchapter C—Aircraft 2017 [24], and [45]). The certification also includes an airworthiness certificate, which is issued when it is demonstrated that an individual aircraft of a given aircraft model is in a condition for safe operation. This refers to the condition of the aircraft relative to wear and deterioration (14 CFR Subchapter C—Aircraft 2017 [24], and [46]). In addition to aircraft, safe aircraft operations also require certification, ratings, privileges, and limitations of airmen, which is based on broad classification of aircraft (14 CFR Subchapter D—Airmen 2017 [24]). Standards have also been established for operating and flight rules (14 CFR Subchapter F—Air Traffic and General Operating Rules 2017 [24]) and the certification of aircraft operators (14 CFR Subchapter G—Air Carriers and Operators for Compensation or Hire: Certification and Operations 2017 [24]). A previous study provided a detailed review of specific classification factors considered in airworthiness standards [52]. This section provides a review at a system level.

2.1.2.1 Federal Aviation Administration (FAA) Aircraft Categorization

The FAA’s aircraft categorization used in these certification standards (14 CFR 2017 [24]) are summarized in Table 6. The categorization first groups aircraft into standard certification, special certification, and certification not required. Aircraft are then further divided into different categories. For each category, the table lists aircraft certification (cert.) standards (14 CFR Parts), aircraft registration requirements (“N-Number,” sUAS “FA-Number” registration, or no registration required), corresponding typical (but not comprehensive and exhaustive) airman certification requirements, operating and flight rules, key aircraft design characteristics, and key operating restrictions.

Table 6: FAA’s Aircraft Categorization in Certification Standards.

Category		Aircraft		Airman Cert.	Operating Rules	Key Design Characteristics	Key Operational Restrictions
		Cert.	Reg.				
Standard Certification	Transport Airplane	§25	N#	Airline transport pilot, aircraft type rating	§121	Multiengine (prop) with passenger seats > 19 or takeoff weight > 19,000 lb, or multiengine (jet) with no established seating or weight minimums; one-engine inoperative climb; increased system redundancy; advanced performance requirements.	Minimum pilot crew ≥ 2 .
	Commuter Airplane	§23	N#	Airline transport pilot, category and class rating	§135	Multiengine (prop); passenger seats ≤ 19 ; takeoff weight $\leq 19,000$ lb	Stalls (except whip stalls); steep turns in which bank angle $\leq 60^\circ$.
	Acrobatic Airplane	§23	N#	Private pilot, single-engine class rating	§91.303	Passenger seats ≤ 9 ; takeoff weight $\leq 12,500$ lb	No maneuver restrictions other than those as a result of required flight tests; not over persons; not over any congested area; not within boundaries of Classes B, C, D, or Class E airspace designated for an airport; within 3 nmi of any Federal airway.
	Utility Airplane	§23	N#	Private pilot, proper class rating	§91		Limited acrobatic: all maneuvers for the Normal category (bank angle $\leq 90^\circ$), plus spins.
	Normal Airplane	§23	N#	Private pilot, proper class rating	§91		Nonacrobatic, e.g., stalls (except whip stalls); lazy eights, chandelles, and steep turns in which bank angle $\leq 60^\circ$.

Category		Aircraft		Airman Cert.	Operating Rules	Key Design Characteristics	Key Operational Restrictions
		Cert.	Reg.				
Standard Certification (Continued)	Very Light Airplane (VLA)	§21.17 (b)	N#	Recreational pilot, single-engine airplane rating	§21.17 (b)	Guidelines in [31], [33]. Engine ≤ 1 (piston); total seats ≤ 2 ; takeoff weight $\leq 1,654$ lb; stalling CAS ≤ 45 kt.	Normal category maneuvers; daylight VFR operations only.
	Transport Rotorcraft	§29	N#	Airline transport pilot, helicopter type rating	§135, §133	Category A: Multiengine with engine and system isolation, and capability for continued safe flight in the event of engine failure; one-engine inoperative climb without ground effect. Required for weight $> 20,000$ lb; passenger seats ≥ 10 .	Radio altimeter (after April 24, 2017);
						Category B: single-engine or multiengine not fully compliant with Category A standards – no guaranteed stay-up ability in the event of engine failure. For weight $\leq 20,000$ lb; passenger seats ≤ 0 , or otherwise provided Category A strength, construction, powerplant, and equipment requirements are met (including one-engine inoperative climb for if passenger seats ≥ 10).	Radio altimeter (after April 24, 2017); takeoff may be begun if the takeoff surface is defined and a landing can be made safely at any point along the flight path if an engine fails;
	Normal Rotorcraft	§27	N#	Commercial pilot, helicopter class rating	§135, §133	Weight $\leq 7,000$ lb; passenger seats ≤ 9 . May be certified as Category A if the specified list of Category A design installation and performance requirements are met.	Radio altimeter (after April 24, 2017); takeoff must be made such that a landing can be made safely at any point along the flight path if an engine fails.
	Manned Free Balloon	§31	N#	Private pilot, balloon rating	§91	Initial rate of climb > 300 ft in the first minute.	None other than general operating and flight rules specified in §91.

Category		Aircraft		Airman Cert.	Operating Rules	Key Design Characteristics	Key Operational Restrictions
		Cert.	Reg.				
Special Certification	Primary Aircraft	§21.24	N#	Private or recreational pilot, proper class rating	§91.325	Unpowered or single engine; total seats ≤ 4 ; unpressurized; weight $\leq 2,700$ lb (land) or 3,375 lb (sea); stall CAS ≤ 61 kt or rotor disc loading ≤ 6 lb/ft ² .	Not for carrying persons or property for compensation or hire; operated by pilot-owner (or a designee) only if aircraft maintained by the pilot-owner.
	Restricted Aircraft	§21.25	N#	As proper	§91.313	Meets airworthiness requirements of an aircraft category except those that are inappropriate for the special purpose, e.g., agricultural, forest, aerial surveying, patrolling, weather control, aerial advertising, etc.	For certificated special purpose only.
	Limited Aircraft	§21.27	N#	As proper	§91.315	Surplus aircraft of the U.S. Armed Forces that is certified in the normal, utility, acrobatic, commuter, or transport category; may be relieved from strict compliance.	Not for carrying persons or property for compensation or hire.
	Provisional	§21 C, I	N#	As proper	§91.317	Aircraft substantially meets/complies with requirement for a type certificate; can be safely operated under appropriate operating limitations.	Not operating aircraft in air transportation.
	Experimental	§21.191	N#	As proper	§91.319	For research and development, certification flight test, air racing, amateur-built, and primary kit-built, etc.	Not for compensation or hire; within an assigned area; not over a densely populated area; daylight VFR operations only.
	Light-Sport Aircraft (LSA)	§21.190	N#	Sport pilot, proper category and class rating	§91.327	Engine ≤ 1 (piston); total seats ≤ 2 ; unpressurized; takeoff weight $\leq 1,320$ lb (land) / 1,430 lb (sea); level CAS ≤ 120 kt; stall CAS ≤ 45 kt; fixed gear (if land); not a helicopter or powered-lift.	Not for compensation or hire; maintained and inspected by a certificated repairman with a LSA maintenance rating.
	Special Flight Permit	§21.197, §21.199	N#	As proper	§21.197	Not currently meeting applicable airworthiness requirements but capable of safe flight.	For ferry, flight testing, evaluating, demo, special operations, etc.

Category		Aircraft		Airman Cert.	Operating Rules	Key Design Characteristics	Key Operational Restrictions
		Cert.	Reg.				
Certification Not Required	Ultralight	No	No	No	§103	Single seat; empty weight < 155 lb (unpowered) or 254 lb (powered); fuel ≤ 5 U.S. gallons; level CAS ≤ 55 kt; and stall CAS ≤ 24 kt.	Recreation or sport only; daylight or civil twilight; visibility ≥ 1, 3, or 5 miles; not over congested areas.
	sUAS	No	FA#, if ≥ 0.55 lb (250 g).	Remote pilot, sUAS rating	§107	Takeoff weight ≤ 55 lb	VLOS; daylight or civil twilight; visibility ≥ 3 miles; not over people; maximum ground speed 87 kt (100 mph); altitude 400 ft AGL or within 400 ft of structure; waivers allowed.
	Model Aircraft	No	FA#, if ≥ 0.55 lb (250 g).	No	§101 E	Weight ≤ 55 lb or otherwise certified by a community-based organization.	Hobby or recreation only; VLOS; operated per safety guidelines of a national community-based organization.

Standard certification categories include *transport airplane, commuter airplane, acrobatic airplane, utility airplane, normal airplane, very light airplane; transport rotorcraft, normal rotorcraft, and manned free balloon*. Each of these categories defines a group of aircraft with distinct design characteristics, intended use, or operating limitations.

Special certification categories include *primary aircraft* [32], *restricted aircraft, limited aircraft, provisional, experimental aircraft, light-sport aircraft (LSA)*, and *special flight permit*. A special certification aircraft category does not necessarily define a group of aircraft with distinct design characteristics. These categories lean more towards using the intended use and operating limitations as a method for categorization. While primary aircraft have similar weight limit and seating limits, they could be airplanes or rotorcraft. The same is true for LSA. Further, a restricted, limited, provisional, or experimental certificate, or a special flight permit may be issued to aircraft that are drastically different in design characteristics, intended use, or operating limitations. Special certification categorization reflects the combined objectives of ensuring operation safety, reducing certification burden, and flexibility to efficiently and effectively manage certification for special situations.

Certification not required applies to three aircraft categories: *ultralight, sUAS, and model aircraft*. The latter two were not introduced until very recently through the FAA's UAS rule making processes [26] [27]. These three categories reflect rule making processes that manage safety by recognizing the smaller, lighter, slower, and simpler aircraft design on one hand, and imposing operating limitations to mitigate risks on the other.

In Table 6, aircraft categories are roughly ordered per the complexity and stringency of the certification process, from the most complex and stringent transport category aircraft at the top, to the simplest and most flexible model aircraft category at the bottom.

2.1.2.2 Observations from FAA's Historical Development

One important observation from a review of the historical development of the FAA's certification standards is that, over time, as technology advances and aircraft design and operating characteristics becomes more diverse for any given category, the certification process for that category becomes more complex. On the other hand, as technology advances, the threshold to access more advanced technology becomes lower. Technologies used to be available to more stringent aircraft categories start to migrate to lower categories. More stringent certification procedures and standards thus should be adopted into the lower aircraft categories. For example, the certification process for both Part 25 (transport) and Part 23 (normal, utility, acrobatic, and commuter) categories have become more complex and burdensome over the years. At the same time, the gap between Part 25 and Part 23 has become smaller.

An upward movement in certification stringency seems unavoidable. This is particularly burdensome for aircraft towards the lower end of the spectrum of a given category. To cope with this issue, a split to an existing, higher aircraft category may be introduced. Aircraft types toward the lower end of the spectrum of a given aircraft category may be downgraded to a lower category, as demonstrated by the introduction of the commuter category into Part 23, which covers smaller multiengine large aircraft that would otherwise be covered by Part 25. Alternatively, aircraft types toward the lower end of a higher category may be introduced as a new, lower category, as demonstrated by the adoption of the VLA category, which covers single-engine airplanes that would otherwise be covered by Part 23 normal category.

Driven by the lower threshold to access advanced technology and market, new, relatively small, simple, and less costly aircraft may also emerge and become popular. New categories may be introduced by recognizing the smaller and simpler aircraft design on one hand, and imposing operating limitations to mitigate risks on the other, as well demonstrated by the introduction of Part 103 ultralight category, the LSA category, Part 107 sUAS category, and the addition of the model aircraft category into Part 101.

2.1.2.3 Part 23 Certification Rule Reform

On August 30, 2017, FAA’s new certification rule for small airplanes [28] becomes effective [40]. The new rule is often referred to as Part 23 rewrite, a process initiated in 2008 with a review of Part 23 [28]. The new Part 23 revolutionizes standards for airplanes weighing 19,000 lb or less and with 19 or fewer passenger seats by replacing prescriptive requirements with performance-based standards coupled with consensus-based compliance methods for specific designs and technologies. These standards also replace the weight and propulsion divisions in small airplane regulations with performance- and risk-based divisions. This regulatory approach recognizes there is more than one way to deliver on safety. It offers a way for industry and the FAA to collaborate on new technologies and to keep pace with evolving aviation designs and concepts. On May 11, 2018, the FAA published a notice of availability accepting 63 means of compliance (MOCs) [29] to Part 23 that are based on consensus standards published by ASTM International. The FAA expects this rule will enable faster installation of innovative, safety-enhancing technologies into small airplanes, while reducing certification costs for the aviation industry.

In the new Part 23, only the normal airplane category is retained. The commuter airplane, acrobatic airplane, and utility airplane categories are deleted. The normal airplane is further categorized using performance- and risk-based divisions. A normal airplane may be certified with 4 difference levels, as shown in Table 7. The normal airplane performance is categorized based on the maximum structural cruising speed (V_{NO}) and maximum operating limit speed/Mach (V_{MO}/M_{MO}), and divided into two levels, as shown in Table 8. The normal airplane may also be certified for aerobatics or not for aerobatics, as shown in Table 9.

Table 7: New Part 23 Normal Airplane Certification Levels.

Certification Level	Definition
Level 1	Airplanes with a maximum seating configuration of 0 to 1 passengers
Level 2	Airplanes with a maximum seating configuration of 2 to 6 passengers
Level 3	Airplanes with a maximum seating configuration of 7 to 9 passengers
Level 4	Airplanes with a maximum seating configuration of 10 to 19 passengers

Table 8: New Part 23 Normal Airplane Performance Levels.

Performance Level	Definition
Low Speed	V_{NO} and $V_{MO} \leq 250$ kt CAS and $M_{MO} \leq 0.6$
High Speed	V_{NO} or $V_{MO} > 250$ kt CAS or $M_{MO} > 0.6$

Table 9: New Part 23 Normal Airplane Aerobatics Categorization.

Aerobatics	Explanation
Not for Aerobatics	Perform any maneuver incident to normal flying, including: stalls (except whip stalls); and lazy eights, chandelles, and steep turns, in which the angle of bank $\leq 60^\circ$
For Aerobatics	Perform maneuvers without limitations, other than those established in the FAA approved flight manual.

Because there is now a single aircraft category in the new Part 23 certification rule. Certification standards are based on the performance-based and risk-based categorization shown in [Table 7](#), [Table 8](#) and [Table 9](#). They are thus forward-looking and flexible to encourage innovation.

2.1.3 Mission, Performance and Equipage-Based Aircraft Categorization

Several performance and equipage-based aircraft categorization methods adopted in manned aircraft operations are of reference value to the sUAS categorization for trajectory modeling purposes. These include flight rules and type of flight, performance-based navigation capabilities, wake turbulence category (WTC), and takeoff and landing performance categorization.

2.1.3.1 Flight Rules and Type of Flight

The FAA transitioned to the new ICAO flight plan form on November 12, 2012 [41]. Item 8 in the new ICAO flight plan form (Appendix 2 in ICAO Doc 4444 [54]) requires the category of flight rules with which the pilot intends to comply. If so required by the ATS authority, the type of flight shall also be provided. Although these two categorizations are for the flight to be executed by the aircraft, they are relevant to the intended flight trajectory. Their selection also depends on the capability of the aircraft and on the flight crew.

ICAO defined flight rules are divided into *visual flight rules* (VFR) and *instrument flight rules* (IFR), as listed in [Table 10](#). An aircraft may not be operated under VFR unless VFR weather minimums are met. ICAO defined types of flight are listed in [Table 11](#) along with reasons for ATS special handling to be entered in Item 18; they imply special types of flights.

Table 10: ICAO Flight Rules.

Code	Category	Explanation
I	Instrument Flight Rules (IFR)	Pilot depends reference to instruments instead of outside visual reference for safe flight.
V	Visual Flight Rules (VFR)	VFR weather minimums are defined by visibility and distance from cloud, depending on the airspace class and altitude.

Table 11: ICAO Type of Flight and Special Handling.

Code		Category	Note
Type of Flight	S	Scheduled air service	
	N	Non-scheduled air transport operation	
	G	General aviation	
	M	Military	
	X	None of the above	
Special Handling	ALTRV	Flight with an altitude reservation	ATFM: Air traffic flow management.
	ATFMX	Approved for exemption from ATFM measures	
	FFR	Fire-fighting	
	FLTCK	Flight check for calibration of navigation aids (NAVAIDs)	
	HAZMAT	Carrying hazardous material	
	HEAD	Flight with Head of State status	
	HOSP	Medical flight declared by medical authorities	RVSM: Reduced vertical separation minimum.
	HUM	Humanitarian mission	
	MARSA	Military aircraft, separation managed by a military entity	
	MEDEVAC	Life critical medical emergency evacuation	
	NONRVSM	Non-RVSM capable flight intending in RVSM airspace	
	SAR	Search and rescue mission	
	STATE	Military, customs or police services	
	Other	Other reasons may be denoted by free text	

2.1.3.2 Equipment and Capabilities

Item 10 in the ICAO flight plan form [54] requires the presence of relevant serviceable equipment on board the aircraft, equipment and capabilities commensurate with flight crew qualifications, and where applicable, authorization from the appropriate authority.

Equipment and capabilities are divided into two main groups: 1) radiocommunication, navigation and approach aid, as listed in [Table 12](#) and surveillance, as listed in [Table 13](#). Because the multiple equipment and capabilities may be available or approved, multiple entries are allowed in the flight plan form. For radiocommunication, navigation and approach aid, the entry could be either N, or S, or multiple entries of individual equipment and capabilities. For surveillance, the entry could be either N or multiple entries of individual equipment and capabilities.

Table 12: ICAO Communication, Navigation and Approach Aid Equipage and Capabilities.

Code	Description	Code	Description
N	No equipment or unserviceable	J5	CPDLC FANS 1/A satellite communication (SATCOM) Inmarsat
S	Standard, including very high frequency (VHF) radiotelephony (RTF), VHF omnidirectional range (VOR) and instrument landing system (ILS)	J6	CPDLC FANS 1/A SATCOM Multifunctional Transport Satellites (MTSAT)
A	Ground-based augmentation system (GBAS) landing system	J7	CPDLC FANS 1/A SATCOM Iridium
		K	Microwave landing system (MLS)
B	localizer performance with vertical guidance (LPV): approach with vertical guidance (APV) with satellite-based augmentation system (SBAS)	L	ILS
		M1	Air traffic control (ATC) satellite voice (SATVOICE) Inmarsat
C	Long range navigation C (LORAN C)	M2	ATC SATVOICE MTSAT
D	Distance measuring equipment (DME)	M3	ATC SATVOICE Iridium
E1	Flight management computer (FMC) waypoint reporting (WPR) aircraft communications addressing and reporting system (ACARS)	O	VOR
		P1	CPDLC required communication performance (RCP) 400
E2	Digital flight information service (D-FIS) ACARS	P2	CPDLC RCP 240
		P3	SATVOICE RCP 400
E3	Pre-departure clearances (PDC) ACARS	P4-P9	Reserved for RCP
F	Automatic direction finder (ADF)	R	Performance-based navigation (PBN)
G	Global navigation satellite system (GNSS)	T	Tactical air navigational aid (TACAN)
H	High frequency (HF) RTF	U	Ultrahigh frequency (UHF) RTF
I	Inertial navigation system (INS)	V	VHF RTF
J1	Controller pilot data link communications (CPDLC) aeronautical telecommunication network (ATN) VHF data link (VDL) Mode 2	W	RVSM approved
		X	Minimum navigation performance specification (MNPS) approved
J2	CPDLC Future air navigation system (FANS) 1/A HF data link (HFDL)	Y	VHF with 8.33 kHz channel spacing
J3	CPDLC FANS 1/A VDL Mode A	Z	Other equipment or capabilities
J4	CPDLC FANS 1/A VDL Mode 2		

Table 13: ICAO Surveillance Equipment and Capabilities.

Code	Description	Code	Description
N	No surveillance equipment	S	Mode S, altitude and ACID
A	Mode A	X	Mode S, neither ACID nor altitude
C	Mode A and Mode C	B1	1,090 MHz ADS-B Out
E	Mode S, aircraft identification (ACID), altitude, and automatic dependent surveillance broadcast (ADS-B) extended squitter	B2	1,090 MHz ADS-B Out and In
		U1	Universal access transceiver (UAT) ADS-B Out
H	Mode S, ACID, altitude, and enhanced surveillance	U2	UAT ADS-B Out and In
I	Mode S, ACID, but no altitude	V1	VDL Mode 4 ADS-B Out
L	Mode S, ACID, altitude, ADS-B extended squitter, and enhanced surveillance	V2	VDL Mode 4 ADS-B Out and In
		D1	FANS 1/A ADS automatic dependent surveillance contract (ADS-C)
P	Mode S, altitude, but no ACID	G1	ATN ADS-C

2.1.3.3 Performance-Based Navigation (PBN) Capabilities

Aircraft operators need to file the PBN capability information, given in area navigation (RNAV) and required navigation performance (RNP) designations, to receive the desired PBN routing. PBN capability is provided in Item 18 using the categorization listed in [Table 14](#).

ICAO Doc 9613 PBN Manual provides practical guidance on PBN implementation [58]. The PBN concept specifies that aircraft RNAV or RNP system performance requirements shall be defined in terms of accuracy, integrity, continuity, and functionality required for the proposed operations in the context of a particular airspace concept. RNAV and RNP are fundamentally similar. The key difference between them is the requirement for onboard performance monitoring and alerting. Compliance with WGS 84 and data quality are integral into PBN. The PBN concept represents a shift from sensor-based to performance-based navigation. Performance requirements are identified in navigation specifications, which also identify the choice of navigation sensors and equipment (as shown in [Table 14](#)) that may be used to meet the performance requirements. This is because each navigation application must be based upon a particular navigation specification and associated navigation aid (NAVAID) infrastructure, which can be different in different airspace concepts.

RNAV and RNP designations (the number following the RNAV or RNP prefix, as shown in [Table 14](#)) refer to the lateral navigation accuracy in terms of total system error (TSE) in nautical miles (nmi), which is expected to be achieved at least 95% of the flight time by the population of aircraft operating within the airspace, route, or procedure. For example, the aircraft is expected to be within the specified value on either side of the desired track centerline for 95% of the flight

time. *Longitudinal accuracy* is defined as the accuracy of distance reporting or the fix location. For RNP operations, the aircraft, or aircraft and pilot in combination, is required to monitor the TSE, and to provide an alert if the accuracy requirement is not met or if the probability that the TSE exceeds two times the accuracy value is larger than 10^{-5} .

Table 14: ICAO Performance-Based Navigation (PBN) Capability.

Descriptor	PBN Specification	Note
A1	RNAV 10 (RNP 10)	APCH: Approach
B1	RNAV 5 all permitted sensors	
B2	RNAV 5 GNSS	AR: Authorization required
B3	RNAV 5 DME/DME	
B4	RNAV 5 VOR/DME	BARO-VNAV: Barometric vertical navigation
B5	RNAV 5 INS or IRS	
B6	RNAV 5 LORANC	IRS: Inertial reference system
C1	RNAV 2 all permitted sensors	IRU: Inertial reference unit
C2	RNAV 2 GNSS	LORANC: LORAN C
C3	RNAV 2 DME/DME	RF: Constant radius arc to a fix
C4	RNAV 2 DME/DME/IRU	
D1	RNAV 1 all permitted sensors	RNP AR APCH: ICAO Doc 9905 [59] specifies that, an RNP AR APCH has a standard RNP 0.3, and minimum RNP 0.1.
D2	RNAV 1 GNSS	
D3	RNAV 1 DME/DME	
D4	RNAV 1 DME/DME/IRU	
L1	RNP 4	
O1	Basic RNP 1 all permitted sensors	
O2	Basic RNP 1 GNSS	
O3	Basic RNP 1 DME/DME	
O4	Basic RNP 1 DME/DME/IRU	
S1	RNP APCH	
S2	RNP APCH with BARO-VNAV	
T1	RNP AR APCH with RF	
T2	RNP AR APCH without RF	

The ICAO PBN Manual [58] points out that since the relationships among the level of collision risk, accuracy, and route spacing or obstacle clearance are generally complex, it is not appropriate to simply assume that the appropriate route spacing (track-to-track) is four times (4 times) the accuracy value, or to assume that the obstacle clearance is two-times (2 times) the accuracy value. Examples include en route continental route spacing of 30 nmi, which is six times (6 times) the RNAV designation of 5 in a high traffic density environment, and 16.5 nmi, which is less than four times (3.3 times), and 18 nmi, which is also less than four times (3.6 times) in a low traffic density environment, for the same direction and for the opposite direction, respectively. In high density terminal areas, 7 nmi route spacing for straight and turning tracks, which is seven times (7 times) the designation for RNP 1, has been derived by a collision risk analysis undertaken by the European Organisation for the Safety of Air Navigation (EUROCONTROL).

Because the desired routing may contain ATS routes with different RNAV/RNP values, multiple entries are allowed in the flight plan form. For helicopters, the ICAO PBN Manual [58] notes that the helicopter community identified the single RNP 0.3 specification for all phases of flight, and that such a single specification would allow a significant part of the IFR helicopter fleet to benefit from PBN. The FAA has developed helicopter guidelines for RNP 0.3 en route and terminal operations in congested airspace to take advantage of the slow speed and high maneuverability of rotorcraft [34].

Some navigation specifications include requirements for vertical guidance, for example barometric VNAV (BARO-VNAV), or altitude constraints for PBN routes. However, these requirements do not constitute vertical RNP, which is neither defined nor included in the PBN concept [58].

2.1.3.4 Wake Turbulence Category (WTC)

Wake turbulence is a byproduct of lift. It is particularly severe behind large jet aircraft. Sufficient in-trail separation must be maintained to avoid the danger of wake turbulence generated by the leading aircraft to the trailing aircraft. The location and strength of wake turbulence are difficult to predict. To determine wake turbulence separation, aircraft have been mainly classified by their maximum takeoff weight (MTOW) or maximum takeoff mass (MTOM) into three categories. FAA aircraft weight classes [43] and ICAO [54] WTC are listed in Table 16. As shown here, there isn't an exact match between the FAA aircraft weight class and the ICAO WTC.

The small number of wake turbulence categories cause unknown, but large, buffers in separation and less efficient use of airspace resources. To handle super heavy jet transport, like Airbus A380-800, the FAA has added the *Super* (J) weight class. In the terminal area, aircraft with MTOW more than 12,500 lb are further categorized as *Small*+. To further improve operational efficiency, the FAA's wake turbulence recategorization (RECAT) [42] program seeks to refine the wake turbulence category by considering wing span in addition to MTOW. The definition of FAA RECAT categories are listed in Table 16.

The FAA and the ICAO assign wake turbulence categories to helicopters and fixed wing aircraft using the same categorization standards, even though it is identified that for the same weight, helicopter wake vortices are more intense than those of fixed wing aircraft [57]. During forward flight, helicopter wake vortices are similar to those of fixed wing aircraft. During hovering or slow taxiing, a helicopter generates strong downwash from its main rotor(s). It is required that a helicopter be kept clear of light aircraft while hovering or air taxiing. Pilots (of either fixed wing

aircraft or helicopters) are advised to avoid taxiing or flying within three rotor diameters of a helicopter hovering or in a slow hover taxiing [35].

Table 15: FAA Aircraft Weight Class and ICAO Wake Turbulence Categories (WTC).

FAA Aircraft Weight Class			ICAO WTC		
Code	Class	MTOW	Code	WTC	MTOM
J	Super	A380-800 and An 225	H	Heavy	$\geq 136,000$ kg
H	Heavy	$\geq 300,000$ lb (136,000 kg)			
L	Large	$< 300,000$ lb (136,000 kg), $> 41,000$ lb (18,600 kg)	M	Medium	$< 136,000$ kg, $> 7,000$ kg
S	Small+	$\leq 41,000$ lb (18,600 kg), $> 12,500$ lb (5,670 kg)			
	Small	$< 12,500$ lb (5,670 kg)	L	Light	$\leq 7,000$ kg

Table 16: FAA Wake Turbulence Recategorizations (RECAT).

Category	Wing Span	MTOW
A	> 245 ft	$\geq 300,000$ lb (136,000 kg)
B	> 175 ft, ≤ 245 ft	
C	> 125 ft, ≤ 175 ft	
D	> 125 ft, ≤ 175 ft	$< 300,000$ lb (136,000 kg)
	> 90 ft, ≤ 125 ft	$> 41,000$ lb (18,600 kg)
E	> 65 ft, ≤ 90 ft	
F	≤ 125 ft	$< 41,000$ lb (18,600 kg)
	Any	$\leq 15,500$ lb (7,031 kg)

2.1.3.5 Aircraft Categorization for Approach and Surface Operations

ICAO Doc 8168 [55] defines typical aircraft performance categories for the development and execution of instrument approach procedures. These categories provide a standardized basis for relating aircraft maneuverability to specific instrument procedures. The dimensions of the aircraft are also a factor for the calculation of the obstacle clearance height. The criterion for this classification is the indicated airspeed (IAS) at the approach end of runway threshold. This speed is equal to the stall speed multiplied by 1.3, or 1-g stall speed multiplied by 1.23 in the landing configuration at the maximum certificated landing mass. If both speeds are available, the higher resulting speed shall be applied. The ICAO aircraft performance categorization is shown in [Table 17](#). For helicopters, the stall speed method does not apply. Where helicopters are operated as

airplanes, they are classified as Category A. When helicopter-specific procedures are used, they are classified as Category H.

Table 17: ICAO Aircraft Approach Speed Categorization.

Aircraft Category	IAS at Threshold	Note
A	< 91 kt	
B	91 kt – 120 kt	
C	121 kt – 140 kt	
D	141 kt – 165 kt	
E	166 kt – 210 kt	
H	Helicopter Specific	

For airport planning and design purposes, the FAA [37] and the ICAO [53] also defined aircraft classification methods based on aircraft dimensions. The FAA classification is called airplane design group (ADG). It considers airplane wing span and tail height. The ICAO classification is called *aerodrome reference code*. It considers the airplane reference field length to determine element 1 of the reference code, and wing span and outer main gear wheel span to determine element 2 of the reference code. The FAA AGD and the ICAO reference code letter are listed in Table 18. The FAA has also adopted the consideration of the dimensions of the aircraft undercarriage to a new classification called *taxiway design group* (TDG). It considers the overall main gear width and the cockpit to main gear distance; and it is presented as polygonal regions in the two-dimensional (2D) space defined by these two parameters. Because taxiway design is less of a concern for sUAS trajectory modeling, details of FAA’s TDG are omitted from the table.

Table 18: Aircraft Classification Methods for Airport Planning and Design.

FAA Airplane Design Group (ADG)			ICAO Aerodrome Reference Code		
ADG	Wing Span (m)	Tail Height (m)	Code Letter	Wing Span (m)	Wheel Span (m)
I	< 15 m	< 6 m	A	< 15 m	< 4.5 m
II	15 m to < 24 m	6 m to < 9 m	B	15 m to < 24 m	4.5 m to < 6 m
III	24 m to < 36 m	9 m to < 13.5 m	C	24 m to < 36 m	6 m to < 9 m
IV	36 m to < 52 m	13.5 m to < 18.5 m	D	36 m to < 52 m	9 m to < 14 m
V	52 m to < 65 m	18.5 m to < 20 m	E	52 m to < 65 m	
VI	65 m to < 80 m	20 m to < 24.5 m	F	65 m to < 80 m	14 m to < 16 m

For the FAA ADG, the most restrictive dimensions of the two determines the category. For the ICAO aerodrome reference code letter, the more demanding code letter between the one that corresponds to the wing span and the one that corresponds to the wheel span determines the category.

2.2 Model Aircraft Categorization

Model aircraft have been playing a vital role in aeronautics research, development, and education. Model flying has been a popular healthy sport for recreation and competition. Model aircraft development pioneered manned aircraft and modern UAS. Local, national, and international model flying communities have been playing an important role in developing, promoting, and administering safety programs that have helped maintain the safety record of model flying. Best practices, procedures, and lessons learned in model flying are valuable assets to the study of sUASs. As such, this section provides a review of existing model aircraft categorization methods.

2.2.1 Definition of Model Aircraft

Model aircraft pioneered the first successful flight of manned aircraft. Both model aircraft and manned aircraft have co-existed since the beginning of manned aviation. However, in the U.S., it was only recently that in Public Law 112-95, “FAA Modernization and Reform Action of 2012” [18], Sec. 336, a formal definition of *model aircraft* was given as:

- (1) capable of sustained flight in the atmosphere;
- (2) flown within visual line of sight of the person operating the aircraft; and
- (3) flown for hobby or recreational purposes.

The FAA Modernization and Reform Action of 2012 prohibits the Administrator of FAA from promulgating any rule or regulation regarding a model aircraft meeting the given criteria. While no other design, construction, performance or operational properties are specified in the definition above, additional criteria that must be met by model aircraft include:

- The aircraft is operated in accordance with a community-based set of safety guidelines and within the programming of a nationwide community-based organization (CBO);
- The aircraft is limited to not more than 55 lb (25 kg) unless otherwise certified through a design, construction, inspection, flight test, and operational safety program administered by a CBO;
- The aircraft is operated in a manner that does not interfere with and gives way to any manned aircraft;
- When flown within 5 miles of an airport, the operator of the aircraft provides the airport operator and the airport air traffic control tower (ATCT) (if applicable) with prior notice.

The above definition and requirements have been adopted by the FAA’s sUAS Final Rule of 2016 (codified in 14 CFR Part 101) [27] and AC 91-57A guidelines [36]. AC 91-57A also provided:

- Limiting operations to within 400 ft AGL as a best practice.

The FAA Modernization and Reform Action of 2012, 14 CFR Part 101, and AC 91-57A all reference to CBO safety programs. Indeed, CBOs have been playing an important role in developing, promoting, and administering safety programs since their very early days, and in maintaining the safety record in the model aircraft community.

The Academy of Model Aeronautics (AMA) is the official national body for model aviation in the United States [3]. In its safety code [4], the AMA defines a *model aircraft* as:

A non-human-carrying aircraft capable of sustained flight in the atmosphere, with limitations including:

- Intended exclusively for sport, recreation, education, and/or competition.
- Not flying higher than approximately 400 ft AGL within three miles of an airport without notifying the airport operator.
- Not exceeding a takeoff weight, including fuel, of 55 lb unless
 - For radio control (RC) airplanes in compliance with the AMA Large Model Airplane Program [6], not exceeding 100 lb (45.4 kg).
- Not using a metal-propellers except
 - For non-aerobatic helicopters operated under metal blade provisions [7], weighing 20 lb or more (but not exceeding 55 lb), and with a rotor diameter greater than 1,800 mm.
- Not using a gas turbine engine unless
 - In compliance with the AMA Gas Turbine Regulations [5], for turbojets and turbofans, total combined installed static thrust not exceeding 50 lb;
 - For control line aircraft, not exceeding static thrust of 20 lb and maximum speed of 100 mph.

Through the National Aeronautic Association (NAA), AMA is recognized by the Fédération Aéronautique Internationale (FAI), the world governing body of all aviation sports activities, as the national organization directing U.S. participation in international aeromodeling activities. The FAI's aeromodeling sporting code [48] identifies model aircraft as Class F aircraft. Specifically, *model aircraft* are defined as:

- An aircraft of limited dimensions, with or without a propulsion device, not able to carry a human being and to be used for contest, sport or recreational purposes.
 - For the whole flight, a RC model aircraft must be within VLOS of the person who directly assumes its control or who is in a situation to take the direct control at any moment, including if the model is being flown automatically to a selected location.
 - For control line model aircraft, the flier must physically hold the control line handle and control the model aircraft himself.
 - Free flight model aircraft must be launched by the flier, and must not be equipped with any device that allows them to be flown automatically to a selected location or controlled remotely during the flight other than to stop the motor and/or to terminate the flight.
- Unless specified otherwise in the rules for a particular class or in the records rules, model aircraft shall meet the following general specifications:

– Maximum flying weight with fuel	25 kg
– Maximum surface area	500 dm ²
– Maximum loading	250 g/dm ²

- Maximum swept volume of piston motor(s) 250 cm³
- Electric motors power source max. no load voltage 72 V
- Maximum total thrust of turbines 25 kg (250 N)
- Noise limits shall be applied to powered model aircraft categories, with 96 dBA at 3 m for any category which does not have approval for any other noise rule. Noise limits do not apply to model aircraft equipped with turbines.

The definitions listed above indicate that the general properties of model aircraft, particularly the maximum flying weight of 55 lb (25 kg), are the same as the FAA's sUAS, but in special cases, a model aircraft may far exceed the basic limits imposed on sUASs. For example, the AMA large scale airplane could weigh 100 lb, and the AMA turbine powered airplane could reach a speed of 200 mph (174 kt), instead of the 100 mph (87 kt) limit. Techniques and design features targeted for competition have provided a foundation for many of today's sUAS designs. Model aircraft categorization methods could thus provide valuable information on how sUAS categorization can be performed.

2.2.2 Model Aircraft Categorization

The FAI model aircraft categorization is primarily based on the methods of remote control, propulsion, and airframe configuration. The complete list of FAI classes is listed in Section 4.B of FAI sporting code [48]. At the top level, model aircraft are divided into the categories listed in [Table 19](#).

The categorization is further refined based on the type of competition, and in some cases vehicle size, weight, and required pilot skill level. Specific design characteristics are imposed to limit and level set aircraft performance among competitors. There is also a division between indoor and outdoor competitions. For the purpose of sUAS operations analysis, control line model aircraft and indoor aircraft are not as relevant as other classes, thus they are ignored in this study. As time progresses and technology advances, new classes of model aircraft may be first introduced as provisional classes, and converted to formal classes later.

FAI classes relevant to sUAS operations are listed in [Table 20](#). This table lists model aircraft characteristics, allowable weather conditions, safety requirements, and noise limitations. In this table, values shown in gray shaded cells are not official requirements. Rather, they are derived from office requirements for reference purposes only.

Main model aircraft characteristics include: task to be completed, propulsion power source, total aerodynamic surface area or disk swept area in case of rotor craft, gross weight with or without fuel/battery/rubber band used for propulsion, aerodynamic surface or disk loading, and automation system. These characteristics determine model aircraft flight characteristics, thus are relevant to trajectory modeling.

Weather conditions are important factors. Specific factors considered by the FAI aeromodelling code include winds, visibility, the position of the sun (which may impact visual observation), and other atmospheric conditions. Except for winds, weather conditions are subjectively evaluated by judges to determine if an interruption of the contest should be initiated. The FAI specifies that winds are measured at 2 m AGL at the start line or model launch area. Wind speed is considered exceeding the allowable limits if the measured wind speed is greater than the given limit for a given duration, e.g., 20 s or 1 min, which are determined for each class separately.

The general limit is 12 m/s at 2 m AGL for more than one minute. Note that the measurement height is different from the current meteorological observation standard of 10 m (typically 33 ft or 27 ft in the United States) AGL [70]. A proper atmospheric boundary layer wind profile must be referenced when interpreting the wind speed limits listed in [Table 20](#).

Table 19: FAI Top Level Model Aircraft Categorization.

Class	Category	Description
F1	Free Flight	Flight during which no physical connection between the model aircraft and the competitor or helper exists. Closed loop control not allowed. Launched by hand.
F2	Control Line Flight	Flight during which all flight controls is accomplished via one or more inextensible wires or cables directly connecting a hand held device and the model aircraft. Automatic flight path control not permitted.
F3	Radio Control Flight	Flight during which the model aircraft is maneuvered by control surface(s) by the flyer on the ground using radio control.
F4	Scale Model Aircraft	Model aircraft as a scaled down replica of a heavier-than-air man-carrying aircraft.
F5	Radio Control Electric Powered Aircraft	Model aircraft powered by electric motors using power packs with no fixed connection to the ground or another model aircraft. Solar cells permitted.
F6	Airports Promotion	Events designed to be the means of promoting aeromodelling sport. Deleted since 2016. Absorbed by other categories.
F7	Radio Control Aerostats	Lighter than air aircraft.
F9	Drone Sport	Multi-rotor drone racing. F9 is introduced in 2019. It was previously listed under F3.

The FAI has established a set of safety rules for model design, test, flying, and contest. Such safety rules may be adapted to a specific class; for example, the specification of flight site safety area and safety line are often class-specific. Subjective determination by judges is also applicable to the safe operations of many model aircraft classes. These safety rules are referred to as basic rules in [Table 20](#). For some classes, specific requirements are in place to ensure safety or to mitigate safety risks. For example, the F3U RC Multi-Rotor First Person View (FPV) Racing class has class-specific requirements related to FPV video link quality for the flight site and class-specific requirements for maneuvers that might lead to a mid-air collision. Such safety rules are referred to as specific rules in the table. For some other classes, particularly F3D Pylon Racing—reflecting the highest level of airframe design and piloting—unique safety rules in aircraft system design and operations are strictly enforced. For example, the F3D contest is designed for average course speed not to exceed 65 m/s (234 km/h). Such safety rules are referred to as strict rules in the table.

Table 20: FAI Model Aircraft Classes Relevant to sUAS Operations.

Class	Category	Task	Power	Wing Span or Diameter	Surface or Disk Area	Weight	Loading	Auto Flight	Wind at 2 m AGL	Safety Rules	Noise at 3 m
F1A	Free Flight Glider	Duration, up to 180 s or 240 s	Hand held 50 m towline		32-34 dm ²	≥ 410 g	> 12 g/dm ²	Not allowed	≤ 9 m/s	Basic	
F1B	Free Flight Wakefield		Extensible, ≤ 30 g (mass)		17-19 dm ²	≥ 200 g ^b	> 12 g/dm ²				
F1C	Free Flight Piston		Piston, ≤ 2.5 cm ² for 4 s		< 37.5 dm ²	≥ 300 g/cm ²	≥ 20 g/dm ²				≤ 6 dBA above ambient at field boundary
F1E	Free Flight Slope Soaring	Duration, up to 120-300 s	Hand launch		≤ 150 dm ²	≤ 5 kg	≤ 100 g/dm ²	Direction			
F1Q	Free Flight Electric Powered Aircraft	Duration, up to 180 s	3 joules per gram of total weight, motor run ≤ 30 s					Not allowed			
F3A	RC Aerobatic	Aerobatic	Any, ≤ 42.56 V if electric	≤ 2.0 m		≤ 5 kg		Not allowed	≤ 12 m/s	Specific, auto power shutoff	≤ 94 dBA
F3S^a	RC Jet Aerobatic		Turbojet or ducted fan, ≤ 42.56 V if electric								
F3M	RC Large Aerobatic		Any	≥ 2.1 m		≤ 25 kg					

Class	Category	Task	Power	Wing Span or Diameter	Surface or Disk Area	Weight	Loading	Auto Flight	Wind at 2 m AGL	Safety Rules	Noise at 3 m
F3B	RC Multi-Task Glider	Duration; distance; speed	Electric winch, 200 m upwind turnaround		≤ 150 dm ²	≤ 5 kg	≤ 75 g/dm ²	Not allowed	≤ 12 m/s	Basic	
F3F	RC Slope Soring	Speed	Hand launch						≥ 3 m/s, ≤ 12 m/s		
F3H ^a	RC Cross Country Soaring	Speed, cross country 20 km course	Electric winch, 300 m upwind turnaround		≤ 150 dm ²	≤ 5 kg	≤ 75 g/dm ²	Any	≤ 12 m/s	Specific, chasing vehicle	
F3J	RC Thermal Glider	Duration	Hand held 150 m towline				12-75 g/dm ²	Not allowed		Basic	
F3K	RC Hand Launch Glider	Duration	Hand launch	≤ 1.5 m		≤ 600 g		Not allowed	≤ 8 m/s	Specific, airspace	
F3C	RC Aerobatic Helicopter	Aerobatic	Any, ≤ 51 V if electric		≤ 250 dm ²	≤ 6.5 kg		Heading Gyro	≤ 8 m/s	Specific, piloting	≤ 89 dBA
F3N	RC Freestyle Helicopter				No limit				Attitude Gyros	≤ 12 m/s	Specific, piloting
F3D	RC Pylon Racing	Speed, 10 laps/4 km course	Piston, ≤ 6.6 cm ²	≥ 1.15, ≤ 1.8 m	≥ 34 dm ²	≥ 2.25 [Ⓕ] , ≤ 3 kg [Ⓖ]		Not allowed	≤ 12 m/s	Restrict,	≤ 96 dBA
									fail-safe shutoff, average course speed ≤ 65 m/s, inspection etc.		

Class	Category	Task	Power	Wing Span or Diameter	Surface or Disk Area	Weight	Loading	Auto Flight	Wind at 2 m AGL	Safety Rules	Noise at 3 m
F4C	RC Scale	Accuracy, maneuver				≤ 15 kg ^c , ≤ 25 kg		Stabilizer or gyros, with or without speed related automatic gain control derived from GPS	≤ 9 m/s	Basic	≤ 96 dBA
F4G ^a	RC Large Scale Aeroplanes					≤ 20 kg ^c , ≤ 25 kg					
F4K ^a	RC Scale Helicopter					≤ 23 kg; ≤ 23 kg ^c if electric					
F5B	RC Electric Glider	Distance, 150 m legs in 200 s; duration, up to 600 s	≤ 42 V, ≥ 450 g, ≤ 1750 W·min		≥ 26.66 dm ² , ≤ 150 dm ²	≥ 1 kg ^c , ≤ 5 kg	12-75 g/dm ²	Stabilizer	≤ 12 m/s	Basic	
F5J	RC Electric Thermal Glider	Duration, up to 600 s	≤ 42 V, 30 s with altitude penalty	≤ 4 m	≤ 150 dm ²	≤ 5 kg					
F5E	RC Solar Powered Aeroplanes	Distance, 150 m legs in 200 s	Area of solar generator ≤ 25 dm ² ; no buffer battery								Stabilizer, GPS logger
F5D	RC Pylon Racing	Speed, 10 laps/4 km course	≤ 21 V, 200-400 g, ≤ 1000 W·min		≥ 26.66 dm ² , ≤ 150 dm ²	≥ 1 kg, ≤ 5 kg	≤ 65 g/dm ²	Stabilizer			Specific, piloting

Class	Category	Task	Power	Wing Span or Diameter	Surface or Disk Area	Weight	Loading	Auto Flight	Wind at 2 m AGL	Safety Rules	Noise at 3 m
F7B	Airships	Time, distance, or target	Any power source that fits model aircraft definition.			≤ 25 kg				Airship should naturally go down if not running. Thrust forced descent allowed.	
F9U ^a	RC Multi-Rotor Drone Racing	Speed	Electric, ≤ 25.5 V	Rotor diameter ≤ 152 mm; axes of all motors must fit within a 330 mm circle.	At least 3 rotors	≤ 1 kg		Stabilizer	≤ 9 m/s	Restrict, fail-safe shutoff, video link, optional LED lights, etc.	

a Provisional class.

b Weigh less extensible motor(s).

c Weight less fuel or power battery pack(s).

Table 21: AMA National Model Aircraft Classes Relevant to sUAS Operations.

Class	Category	Task	Power	Wing Span	Surface Area	Weight	Loading	Auto Flight	Speed	Wind at 10 m AGL	Safety Rules	Noise at 3 m
	Park Flyer		Electric, rubber, or other quiet motor			≤ 2 lb			≤ 60 mph	≤ 40 mph	Basic	Quiet
755B	RC Combat	Combat	≤ 0.30 in ³			≤ 3.5 lb ^b , ≤ 4.0 lb ^b					Specific, remote shut off	
LMA-1	RC Large Airplane		Non-turbine			55-77.2 lb	≤ 80 oz/ft ²				Restrict, AMA pilot qualification, AMA aircraft certification, redundancy, fail-safe auto retard, remote shut off	
LMA-2						77.3-125 lb						
	RC Turbine		Turbine, thrust ≤ 50 lb			≤ 55 lb			≤ 200 mph			
LTMA-1	RC Large Turbine Model Airplane		Turbine, thrust ≤ weight + 5 lb, but ≤ 90 lb			55-77.2 lb	≤ 100 oz/ft ²		≤ 200 mph			≤ 96 dBA
LTMA-1						77.3-100 lb			≤ 140 mph			

a Weight less fuel.

b Weight with power battery pack(s).

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In order to attract more participants in model aircraft competitions, National Airsport Control (NAC) is encouraged by the FAI to establish further national model aircraft classes in addition to FAI classes. AMA national classes relevant to sUAS operations are listed in [Table 21](#).

One unique AMA class is the Park Flyer model [\[8\]](#), [\[9\]](#). These aircraft are situated towards the lower end of model aircraft in terms of weight and speed, and consequently have lower risks upon impact with properties or persons on ground. One of the intents of the AMA Park Flyer Program is to establish new fields for park pilots in urban as well as rural areas. Such fields may include parks, parking lots, playgrounds, ballfields etc.

RC Combat involves at least four model aircraft flying against each other in a five-minute round, provided the number of pilots allows and no frequency conflicts exist [\[2\]](#). A pilot scores by using its propeller to cut streamers towed behind competing aircraft. RC Combat is thus a challenging competition for the contestants, and is also exciting and enjoyable for spectators around the flying field. As such, specific safety rules are in place to ensure safe competition.

Under the AMA Large Model Airplane Program [\[6\]](#) and AMA Gas Turbine Regulations [\[5\]](#), strict safety rules and procedures have been established. Under these rules and procedures, AMA pilot qualification and AMA aircraft certification are required for operating RC Large Airplanes, regular RC Turbine model aircraft, and RC Large Turbine Model Airplanes. These models are situated towards the upper end of model aircraft in terms of weight and speed, and thus have high risks upon impact with properties or persons on the ground. The restrictive safety rules are essential for the safe operation and healthy development of these model aircraft.

2.3 Existing Unmanned Aircraft System (UAS) Categorization

Various unmanned aircraft categorization schemes have been developed by different regulatory, military, industry, and public organizations. Although many of these categorization schemes cover the broad spectrum of unmanned aircraft in terms of size, performance, and technology, recent fast growing developments in the sUAS arena have drawn attention from regulatory authorities, industry, and the research community to proposing new categorization schemes. Particularly, focus is given to the characterization of risk levels. This section provides a review of major unmanned aircraft categorization methods, including both existing ones (historical and current), and those still under development. This review is not intended to be a complete survey of all UAS categorization methods ever used or studied. Reviews of additional existing UAS categorization methods can also be found in [\[51\]](#), [\[64\]](#), and [\[69\]](#).

2.3.1 Federal Aviation Administration (FAA) UAS Categorization

Although the FAA considers model aircraft as UASs, they are currently operated per safety guidelines of a national community-based organization, as discussed in [Section 2.2](#). Thus, they are not included in the discussion in this section. FAA's existing UAS categorization methods discussed in this section include those presented in official FAA materials, and those considered by FAA chartered committees such as aviation rulemaking committees (ARCs) and Radio Technical Commission for Aeronautics (RTCA) committees.

The simplest UAS categorization is by gross weight (similar to MTOW for manned aircraft). The FAA's UAS categorization by weight is summarized in [Table 22](#). The main sources of this information include FAA Order 7360.1B [\[43\]](#) and recent sUAS final rules [\[26\]](#), [\[27\]](#) and proposed

rule on operation over people [30]. Categorization by weight do reflect UAS performance and systems complexity in some way, but it does not provide a precise grouping of these two parameters, nor is it able to comprehensively capture differences in UAS operational characteristics.

Table 22: FAA UAS Categorization by Weight.

Category		Weight	Note
Other UAS		MTOW > 330 lb (150 kg).	Not officially defined.
Micro-/Ultra-Light UAS		MTOW > 55 lb (25 kg) but ≤ 330 lb (150 kg).	See FAA Order 7360.1B [43].
sUAS	Regular	MTOW > 0.55 lb (250 g) but ≤ 55 lb (25 kg).	See sUAS operation and certification final rule [27].
	Under 0.55 lb	MTOW ≤ 0.55 lb (250 g).	See sUAS registration final rule [26] and proposed rule on operation over people [30].

As a historical account, RTCA DO-304, guidance material and considerations for UAS [74], provided a notional categorization scheme for applying different requirements among the various operations and systems. It was similar to the categorization applied to manned aircraft for certification purposes. Factors considered included the type of operation, aircraft and system, operational environment, and operational complexity. The notional categorization is summarized in Table 23. In this notional categorization, regulatory requirements on detect, sense, and avoid (DSA) capabilities are specifically stated.

Table 23: RTCA DO-304 Notional UAS Categorization.

Category	Description	Operational Characteristics	Regulations
A	Privately owned; for recreational or sport purposes.	Weight limitations similar to model aircraft; VLOS; confined operational areas and altitudes.	Unregulated; guided by the FAA and professional organization(s).
B	Small UAS for non-recreational purposes.	Kinetic energy limitations; fail-safe mechanisms; VLOS; outside the confines of Category A.	Minimal regulation for systems and their pilots; additional guidelines or restrictions by professional organization(s).
C	UAS that does not meet all of Category B restrictions.	Aircraft too large or too fast to meet Category B kinetic energy limitations; BVLOS; restricted in specialized purposes and operating environments.	Detect, sense and avoid (DSA) required for some operations; pilot license required and must comply with Part 91.
D	UAS allowed to “file and fly” similar to manned aircraft.	Equal access to the NAS.	Approved detect, sense and avoid (DSA) required for all operations; conform to certification standards derived from 14 CFR and tailored to UAS operations.

Later, in RTCA DO-320, operational services and environmental definition (OSED) for UAS [75], a categorization was provided per grouping by UAS performance. Five manned aircraft performance categories were adopted: *turbojet fixed-wing*, *turboprop fixed-wing*, *reciprocating/electric fixed-wing*, *VTOL*, and *airship*. These UAS categories are summarized in Table 24. Note that even though climb rate is an important component of the aircraft performance, there is no prescribed climb rate criteria for the categorization.

Table 24: RTCA DO-320 UAS Categories.

Category	Description
Turbojet Fixed-wing	High performance, single- or multi-engine fixed-wing aircraft typically capable of the greatest speeds and climb rates as well as high altitude cruise
Turboprop Fixed-wing	High-performance single- or multi-engine aircraft capable of medium speeds, but rapid climb and descent rates.
Reciprocating or Electric Fixed-wing	Single- or multi-engine aircraft exhibiting slow to moderate speeds, slow to moderate climb and descent rates, majority below 12,000 ft
VTOL	Aircraft capable of takeoffs and landings from a point without the aid of an independent launch or recovery mechanism, and tend to have slow to moderate cruise speeds and climb rates.
Airship	Engine-driven lighter-than-air aircraft that can be steered, and have very limited speed, maneuvering, and climb/descent rates when compared to other aircraft.

Within each of these categories, four additional subcategories were defined: *standard*, *non-standard small*, *non-standard high altitude long endurance (HALE)*, and *conversion*. These subcategories are summarized in Table 25.

Table 25: RTCA DO-320 UAS Subcategories.

Subcategory	Description	Rationale
Standard	Aircraft resembling manned aircraft in terms of size, capabilities, and performance.	To allow comparisons with UAS that are like manned aircraft, and would most likely share altitudes and air traffic services similar to manned aircraft.
Non-standard Small	Aircraft with a physical size and weight that is considerably smaller than the smallest manned aircraft.	Representation of a very large portion of the UAS market.
Non-standard HALE	Aircraft capable of operating at very high altitudes and long endurance well beyond those of manned aircraft.	Low quantities but with extreme performance thus needs to be distinguished.
Conversion	Aircraft are converted from manned aircraft to operate as UAS.	Same as the Standard subcategory, potential large numbers to take advantage of a certification basis.

In 2016, the FAA chartered a micro small UAS (μ UAS) ARC to develop a performance-based standard that would allow micro UAS to be operated over people who are not operating the UAS or under the cover of a structure [10]. In furtherance of that goal, the ARC identified four μ UAS categories, defined primarily by level of risk of injury posed, for operations over people. As a historical reference, these recommended μ UAS categories are summarized in Table 26. The risk considered is the one associated with the UAS directly falling down on people. The risk by lateral impacts due to speed is already addressed in Part 107. The level of risk is defined by the probability of Abbreviated Injury Scale (AIS) [1] level 3 (serious) or greater injury upon impact by the aircraft, measured by impact kinetic energy per unit area. The categorization is completed by UAS performance standards and operational restrictions beyond what are already specified in Part 107. In-depth analyses were carried out by MITRE [71] and others in this area.

Table 26: Micro UAS ARC Recommended Categorization for Operations over People.

Category	Level of Risk	Performance Standards	Restrictions over Part 107
1	Takeoff weight $\leq$ 0.55 lb (250 g); 1% chance of AIS level 3 or greater injury.	No.	No.
2	1% chance of AIS level 3 or greater injury upon impact with a person.	Impact energy, e.g., 12 J/cm ² (e.g., quadcopter of 4-5 lb); industry consensus performance standards for Category 2.	Set-off distances $\leq$ 20 ft above people's heads or $\leq$ 10 ft laterally away from people.
3	30% chance of AIS level 3 or greater injury upon impact with a person.	Impact energy TBD; industry consensus performance standards for Category 3.	Not over crowds; in a closed- or restricted-access work site; overflight of people is limited to those who are transient or incidental.
4	Same as Category 3.	Same as Category 3.	Over crowds or dense concentrations of people; in compliant of consensus-based risk mitigation standards.

On February 13, 2019, FAA published a notice of rulemaking on sUAS over people [30]. In the proposed amendment to Part 107, FAA proposes three categories of permissible routine operations over people without a waiver or exemption, as summarized in Table 27. For Category 2 and Category 3 operations over people, performance standards, and in the case of Category 3, operational restrictions beyond those imposed by the new proposed part 107 are specified. These standards are based on the risk of human injury from a sUAS upon impact. The sUAS must not be capable of causing an injury more severe than that would result from a total impact kinetic energy transfer of 11 ft-lb (Category 2, i.e., low probability of an AIS level 3 injury) or 25 ft-lb (Category 3, i.e., low probability of an AIS level 6 injury) from a rigid object. The sUAS must not have exposed rotating parts that could lacerate human skin and must not have any safety defects identified by the FAA that may cause AIS level 3 (Category 2) or 6 (Category 3) injury. Although airworthiness certification is not required for Part 107 sUAS, the FAA proposes to rely on a declaration of compliance as a document a manufacturer submits to certify that a sUAS conforms

to the Category 2 or 3 requirements for operations over people. The manufacturer must show the means of compliance would achieve the safety level the proposed standards reach. The means of compliance must be accepted by the FAA. The means of compliance may be manufacturer custom or proprietary, voluntary consensus standards, or those established by the FAA.

Table 27: FAA Proposed Categories of Permissible Operations over People.

Category	Performance-based Requirements				Pilot Requirements
	Weight	Injury upon Impact	Exposed Rotating Parts	Safety Defects	
1	≤ 0.55 lb	Not required	Not required	Not required	Ensure aircraft weight.
2	> 0.55 lb, ≤ 55 lb	No more severe than from 11 ft-lb rigid object impact (AIS level 3)	No exposed rotating parts that could lacerate human skin upon impact	No safety defects that may cause AIS level 3 injury.	Ensure aircraft is labeled for Category 2 operations.
3	> 0.55 lb, ≤ 55 lb	No more severe than from 25 ft-lb rigid object impact (AIS level 6)		No safety defects that may cause AIS level 6 injury.	Ensure aircraft is labeled for Category 3 operations. Operations must not over a moving vehicle or over open-air assemblies; over a closed- or restricted-access site with notification, or not maintain sustained flight over people.

The FAA is cognizant of the importance of various stakeholders to be able to identify sUAS to mitigate security concerns that operations of sUAS may present. As a result, the FAA plans to finalize its policy concerning remote identification of sUAS—by way of rulemaking, standards development, or other activities that other federal agencies may propose—prior to finalizing the proposed changes to the Part 107 rule that would permit operations of sUAS over people [30].

The FAA Traffic Alert and Collision Avoidance System (TCAS) Program Office is developing a family of advanced Airborne Collision Avoidance System (ACAS) system, whose original version, ACAS X, is designed as a successor to TCAS. A variant of the system designed for UAS, ACAS Xu, is intended to be a complete detect and avoid (DAA) solution. The ACAS Xu for sUAS (ACAS sXu) concept is an extension of the ACAS Xu concept, and intends to provide a DAA capability for sUAS with various equipage and capabilities and operating under Part 107 or under a waiver. The ACAS sXu is envisioned to provide remain well clear (RWC) protection volumes against manned aircraft and large UAS intruders, and collision avoidance (CA) protection against other sUAS intruders. In defining the ACAS sXu concept of use [38], [39], the sUAS equipage is divided into three example classes. Table 28 is a summary of the FAA sUAS DAA equipage classification based on the interpretation, but not an exact representation, of the ACAS sXu concept. Nonetheless, the summary in Table 28 provides a reference to the types of equipage and capability categorization that may be considered for sUAS operations.

Table 28: FAA Example sUAS DAA Equipage Classification.

Class	Airborne Surveillance	Physical Architecture
I	ADS-B In Non-cooperative sensor(s) Small UAS local broadcast remote tracking Out/In Cloud-enabled air-ground link Active interrogator	ACAS sXu on board Optional ground surveillance uplink Auto onboard or remote response
II	ADS-B In Small UAS local broadcast remote tracking Out/In Cloud-enabled air-ground link	ACAS sXu on board Ground surveillance uplink Auto onboard or remote response
III	None	ACAS sXu on ground Telemetry downlink Remote response

2.3.2 Department of Defense (DoD) UAS Categorization

The Department of Defense (DoD) has been using a UAS categorization that identifies UAS by attributes of weight, operating altitude, and airspeed into five groups [19]. These groups are summarized in [Table 29](#).

Table 29: Department of Defense (DoD) UAS Groups.

Group	MTOW	Operating Altitude	Speed	Operational Environment
1	0-20 lb	< 1,200 ft AGL	100 kt	Typically hand-launched, portable systems
2	21-55 lb	< 3,500 ft AGL	< 250 kt	Usually from unimproved areas and via catapult
3	< 1,320 lb	< Flight Level (FL) 180		Usually from unimproved area, may not require an improved runway
4	> 1,320 lb	> FL 180	Any airspeed	Normally from improved areas, beyond line of sight (BLOS) communications, and stringent airspace operations requirements
5				

The DoD UAS categorization has been adopted by several U.S. armed forces including the Army [12] and the Marine Corps [66]. The U.S. Air Force (USAF) recognizes the DoD UAS groups [76], [77], but it further classifies aircraft of the size of MQ-1B Predator (a Group 3 UAS) or larger as a Remotely Piloted Aircraft (RPA), while aircraft smaller than 1,320 lb (Group 3 or lower) as a Small UAS (SUAS). Note that the definition of USAF SUAS is different from the FAA sUAS definition. The former has a much wider weight range, covering Groups 1-3, while the latter covers Groups 1 and 2 only. The USAF intends to adopt the FAA definition in the future.

2.3.3 United Kingdom (UK) UAS Categorization

In the United Kingdom (UK), the Ministry of Defence (MOD) recognizes the importance of a classification system and that, given the diversity of unmanned aircraft and their capabilities, there is no easy one-size-fits-all classification system [67]. For example, the 30 lb Laima unmanned aircraft from Aerosonde had demonstrated transatlantic capability in 1998. As part of the UK's policy to implement North Atlantic Treaty Organization (NATO) doctrine, the UK MOD had previously adopted the NATO unmanned aircraft classification, while acknowledging that a minor update may be required in the future. This classification from these documents are summarized in Table 30. It uses MTOW as a basic criterion to create 3 classes, each of which may be subdivided, if required, to create categories including *micro*, *mini*, *small*, *tactical*, *medium altitude long endurance* (MALE), *HALE*, and *strike/combat*.

Table 30: UK Ministry of Defence (MOD) Unmanned Aircraft Classification.

NATO Class	Category	Normal Employment	Normal Operating Altitude	Normal Mission Radius	UK Civil Category
Class I < 150 kg	Class I(a) Nano < 200 g				
	Class I(b) Micro < 2 kg	Tactical Platoon, Section, Individual (single operator)	Up to 200 ft AGL	5 km (Line of sight (LOS) communications)	Weight Classification Group (WCG) 1 Small Unmanned Aircraft < 20 kg
	Class I(c) Mini 2-20 kg	Tactical Sub-Unit (manual launch)	Up to 3,000 ft AGL	25 km (LOS)	
	Class I(d) Small > 20 kg	Tactical Unit (employs launch system)	Up to 5,000 ft AGL	50 km (LOS)	WCG 2 Light Unmanned Aircraft 20-150 kg
Class II 150-600 kg	Tactical	Tactical Formation	Up to 10,000 ft AGL	200 km (LOS)	WCG 3 UAV > 150 kg
Class III > 600 kg	MALE	Operational/ Theatre	Up to 45,000 ft AGL	Unlimited (BLOS)	
	HALE	Strategic/ National	Up to 65,000 ft AGL		
	Strike/ Combat				

Table 30 also lists normal employment, operating altitude, and mission radius where applicable. For normal mission radius, it is also indicated if line of sight (LOS) or beyond line of sight (BLOS) communications are required.

In early 2015, MOD's Military Aviation Authority (MAA) published a modified remotely piloted aircraft system (RPAS) categorization [62]. It is said to be scalable, allowing greater

flexibility in the application of regulations. The NATO classification, which has added a Nano Category (< 200 g), has been utilized as the baseline. The UK classification has been that has been formally adopted in Regulatory Article 1600 [63], and [68] that replaced [67]. MTOW is no longer used as the sole determinant of the final categorization of an RPAS. It is considered alongside risk aggravating and mitigating factors of its operation and characteristics. The modified RPAS categorization philosophy is shown illustratively in Figure 3 ([62], [63]). Example mitigating factors may include operation in VLOS, in segregated airspace, and over low population density areas; flight termination system; redundancy; and frangibility of RPAS structure. Example aggravating factors may include extended range BVLOS, in non-segregated airspace, and over congested or high population density areas; weaponization; failure model or high kinetic energy; and complexity.

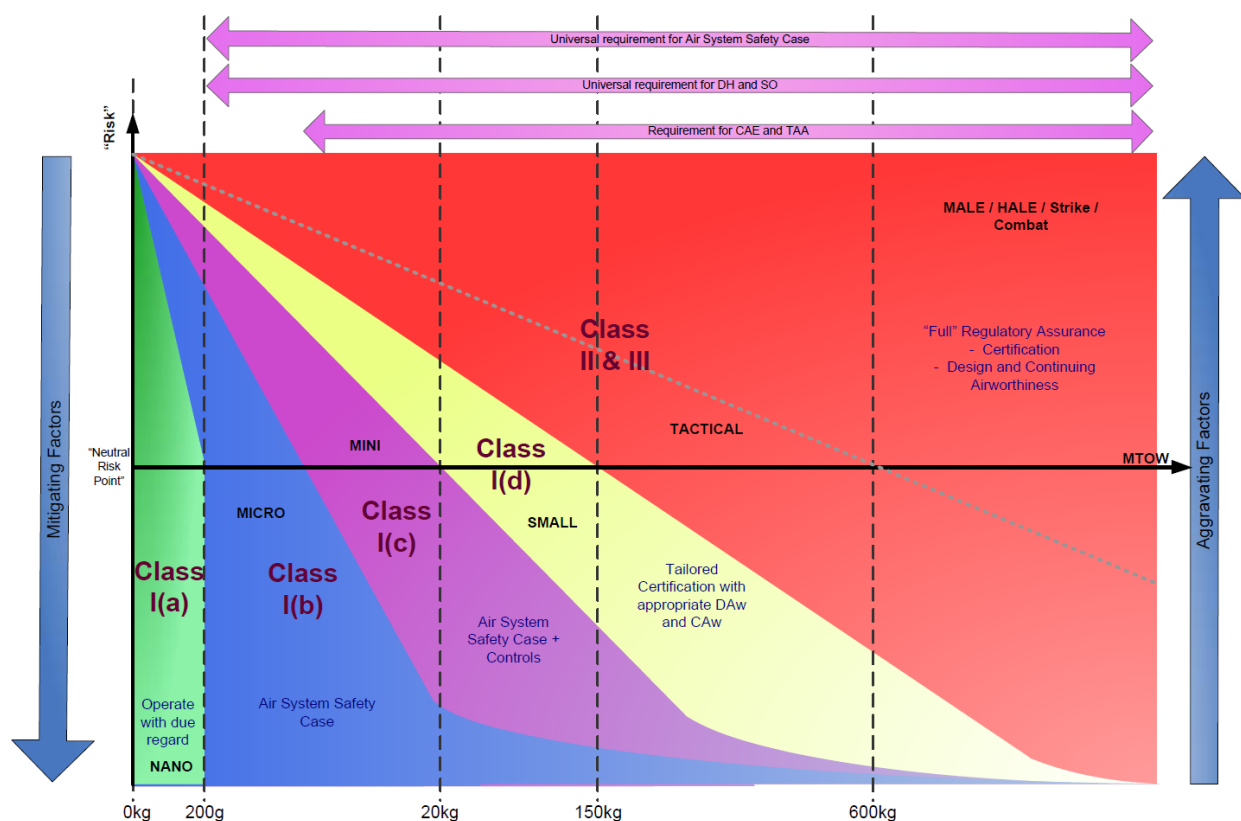


Figure 3: United Kingdom (UK) Mistry Aviation Authority (MAA) remotely piloted aircraft system (RPAS) categorization philosophy.

The UK Civil Aviation Authority (CAA) currently uses a simple UAS classification with zero fuel mass (ZFM) as the main discriminant [13]. Using this classification, UASs are divided into small unmanned aircraft (SUA), light UAS, and regular UAS, as shown in Table 31. In CAA's air navigation order (ANO) [15] (with relevant amendments as presented in [14]) a threshold of ZFM 7 kg is also used in certain rules. Additional rules also apply to small unmanned surveillance aircraft (SUSA)—SUA equipped to undertake any form of surveillance or data acquisition. These items are also listed in Table 31.

The CAA's UAS concept of operations (ConOps) [13] further describes an approach taken by the CAA to determine the level of assurance and assessment required prior to the issuance of a permission. This approach categorizes UAS and the intended operations into either Category A, B,

or C, as shown in Figure 4. Category A may be extended for a higher mass, depending on the environment and complexity of the system, e.g., to 20 kg. In the figure, *Technical Complexity* refers to how complex the system is (e.g., number of flight control modes, flight management systems, etc.); whereas *Operating Environment Complexity* refers to how complex the environment is (e.g., congested areas, complex airspace, etc.).

Table 31: UK Civil Aviation Authority (CAA) Classification.

Category	Zero Fuel Mass		Operating Conditions		Regulatory Body
SUA	0-7 kg	SUA	Permission required for commercial operations; without complying with airworthiness or flight crew licensing requirements.	VLOS, 500 m from the pilot.	CAA
		SUSA		Same as above, and: permission required for over or within 150 m of congested area or open-air assembly of more than 1,000 persons; or within 50 m of any vessel, vehicle or structure, or any person not under the control of the remote pilot; during takeoff or landing, within 30 of any person not under the control of the remote pilot.	
	> 7 to 20 kg	SUA	Operating height or distance ≤ 400 ft AGL or from surface.	Same as 0-7 kg SUA, and: ATC permission required in controlled airspace or aerodrome traffic zone.	
		SUSA		Same as above, and requirements for 0-7 kg SUSA.	
Light UAS	> 20 to 150 kg		Must comply with airworthiness or flight crew licensing requirements unless with an exemption.		
UAS	> 150 kg		European Aviation Safety Agency (EASA)		

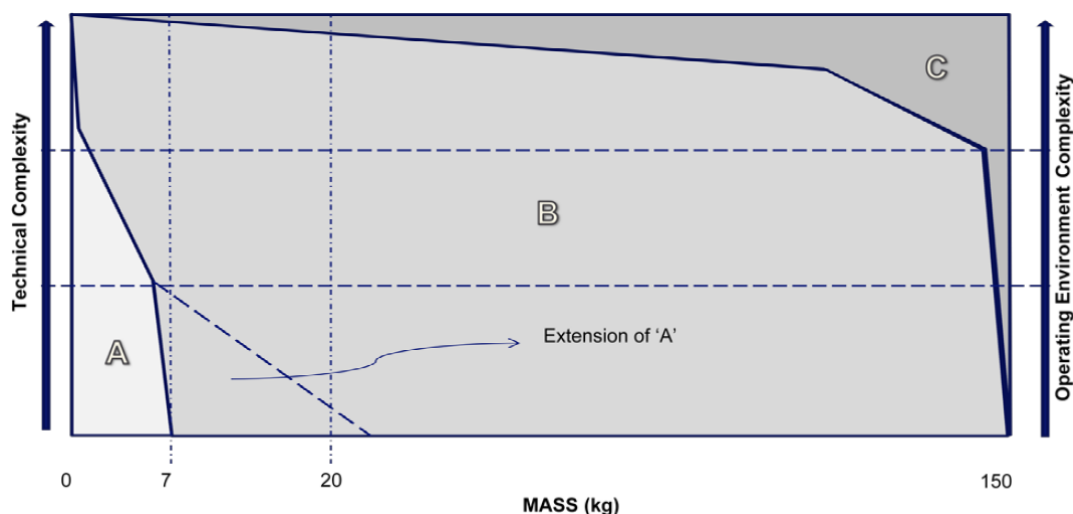


Figure 4: UK Civil Aviation Authority (CAA) UAS Categorization Map.

The criteria used by CAA in issuing an exemption or permission are summarized in [Table 32](#). The CAA airworthiness assessment is based on a UAS Operating Safety Case (OSC) submission that follows templates divided into three separate volumes. The CAA has defined an *extended visual line of sight* (EVLOS) as either within or beyond 500 m horizontal distance from the remote pilot, where direct visual contact is addressed via other methods or procedures.

Table 32: UK CAA UAS Approval Criteria for Operation Categories.

Category	A	B	C
Airworthiness	No airworthiness approval or TC.	Airworthiness assessment-based upon OSC; no design approval or TC.	TC Basis agreed; possibility of TC; design and production approval required.
Operational	No operational approval required or approval based on OSC Volume 1.	Approval based on OSC Volume 1, 2, and 3 submission.	Approval based on OSC and airworthiness submission.
Pilot Competence	For standard permission, evidence of pilot competency.	Evidence of pilot competency.	Evidence of pilot competency.
Operating Environment	VLOS; not in densely populated areas; ≤ 400 ft AGL; very low risk environment.	Congested areas; complex environment; VLOS, EVLOS.	Congested areas; and/or high risk environment; and/or BVLOS, BLOS; and/or complex airspace.
Influencing Factors	Low complexity; low mass; benign operating environment.	Low to medium complexity; congested areas or higher risk environment; increased mass.	High complexity; and/or high mass; and/or densely populated or high risk environment; over groups of people; complex flight profiles; swarming or multiple UAS under one operator.

2.3.4 European Union Existing UAS Categorization

For regulation purposes, the European Aviation Safety Agency (EASA) is proposing to establish three categories of operations and their associated regulatory regimes [20], [21], [22], as summarized in [Table 33](#). This is the same categorization being adopted by the Joint Authorities for Rulemaking of Unmanned Systems (JARUS), except that categories *Open*, *Specific*, and *Certified* are codified by JARUS as Categories A, B, and C, respectively.

Considering the different levels of risk within an Open category operation, this category is further divided into subcategories. Each subcategory is characterized using the specific class of UAS, area of operation, and remote pilot competency. The UAS classes are mainly defined by MTOM or kinetic energy, along with technical requirements and electronic identification (ID) and geo awareness (geo fencing) requirements, as shown in [Table 34](#). Subcategories under the Open category are summarized in [Table 35](#). Examination indicates that the EASA-proposed Open category is roughly equivalent to the FAA routine Part 107 operations in terms of aircraft size.

However, the EASA proposed Open subcategories are somewhat more restrictive than the FAA proposed categories for operations over people, and lack some of the flexibilities allowed by the FAA proposed rule for the manufacture to incorporate a wide variety of means to comply with the performance-based impact energy transfer standards.

Table 33: European Aviation Safety Agency (EASA) UAS Operation Categorization.

Category	Level of Risk	Safety Assurance	Regulation and Authorization
Open	Low Risk	Safety is ensured through operational limitations, compliance with industry standards, requirements on certain functionalities, and a minimum set of operational rules, specifically: MTOW < 25 kg, < 120 m AGL, and VLOS.	Does not require a prior authorization before the operation takes place.
Specific	Medium Risk	Authorization by national aviation authorities (NAAs) following a risk assessment performed by the operator. A manual of operations shall list the risk mitigation measures.	Requires an authorization before the operation takes place based on specific operation risk assessment (SORA).
Certified	Higher Risk	Requirements comparable to manned aviation requirements. Oversight by NAAs and by EASA (design and approval of foreign organizations).	Requires the certification of the aircraft, a licensed remote pilot, and an operator approved by authority.

Table 34: EASA UAS Classification.

Class	MTOM or Joule	Technical Requirements	Electronic ID/ Geo Awareness	UAS Registration
C0	< 250 g	Consumer information, Toy Directive or 19 m/s, no sharp edges, selectable height limit	No	No
C1	< 80 J or < 900 g	Consumer information, 19 m/s, kinetic energy, mechanical strength, lost-link management, no sharp edges, selectable height limit	Yes, unique serial number as ID	Yes
C2	< 4 kg	Consumer information, mechanical strength, lost-link management, no sharp edges, selectable height limit, frangibility, low-speed mode		
C3	< 25 kg	Consumer information, lost-link management, selectable height limit, frangibility		
C4		Consumer information, no automatic flight	If required by zone of operations	

Table 35: Subcategories under the EASA UAS Open Category.

Subcategory	Area of Operation	Remote Pilot Competency	UAS Class
A1 Fly over People	Over uninvolved people but not over crowds	Consumer info	C0
		Consumer info Online train and test	C1
A2 Fly Close to People	A safe distance from uninvolved people	Consumer info Online train and test In person theoretical test	C2
A3 Fly Far from People	Area where no uninvolved people will be endangered, and safe distance from urban areas	Consumer info	C3
		Online train and test	C4

For RPAS integration, EUROCONTROL is using a traffic classification [23] for very low altitude (below 500 ft) operations. This is because, in EUROCONTROL’s view, RPAS are very difficult to categorize due to the large variety of shapes, sizes, and performance. A class of RPAS traffic is a set of flying rules, operational procedures, and system capabilities applicable to the RPAS and to the operator when operating the RPAS in a portion of the airspace. The EUROCONTROL RPAS traffic classification consists of four very low altitude classes (below 500 ft) and three classes for operations up to above FL600, as summarized in [Table 36](#).

2.3.5 Civil Aviation Administration of China (CAAC) UAS Categorization

In Civil Aviation Administration of China (CAAC) interim regulations for light and small UAS operations [16] and civil UAS pilot regulations [17], a UAS categorization is provided, as summarized in [Table 37](#). This categorization considers aircraft weight and operational characteristics. It provides a relatively simple framework for specifying operational and pilot license requirements. The CAAC interim light and small UAS regulations apply to the UAS categories listed in [Table 37](#), but they do not apply to model aircraft, unless they are installed with an autopilot, command and control data links, or autonomous flight equipment, nor do they apply to indoor operations.

The CAAC has authorized commercial cloud-based UAS services (UAS Cloud) to serve as active surveillance systems. For certain UAS categories, either a direct aircraft to UAS Cloud connection or a ground station to UAS Cloud connection is required for the operator to send position, altitude, and speed information at report rates ranging from once per second to once per minute, depending on the area where the UAS is operated. For Category VI, which has an MTOW exceeding what defined for an FAA sUAS but matching an FAA micro-/ultra-light UAS, passive surveillance equipment is required. *Passive surveillance system* refers to radar, automatic dependent surveillance broadcast (ADS-B) or the BeiDou Navigation Satellite System (BDS), all of which allow information to be sent from the aircraft directly to surveillance systems without passing through the UAS operator’s ground system.

Table 36: EUROCONTROL RPAS Traffic Classification.

Class	Description	EASA Category	VLOS or BVLOS	Operational Requirements
I	Buy and Fly	Open	VLOS	In low risk environments; capable to self-separate in 3D; using geofencing to remain clear of no-drone zones like airports; mandatory declaration.
II	Free Flight	Specific or Certified	VLOS and BVLOS	Free flight path with no fixed route structure; up to 500 ft AGL; capable to self-separate in 3D; surveillance capability; barometric equipage if BVLOS; mandatory authorization.
III	Free Flight or Medium/Long Haul Route Structure	Specific or Certified	BVLOS	Free flight or within a route structure; up to 500 ft AGL; surveillance capability; barometric equipage; mandatory authorization; mainly for medium/long haul transport.
IV	Special Operations	Specific or Certified	VLOS and BVLOS	Designated for highly specialized operations; up to 500 ft AGL; could require surveillance capability; special authorization on case by case basis; could be civil, state or military operations.
V	IFR/VFR Operations outside ATS Network		VLOS and BVLOS	Outside ATS route network and segregated launch and recovery at airports; up to FL600; no additional performance requirements compared to manned aircraft; RPAS two-way communications with ATC if required; operator able to contact ATC regarding special conditions; DSA; flight plan required.
VI	IFR Operations including within Network		BVLOS	Including ATS route network, terminal and airports; up to FL600; either manned transport enabled to fly unmanned or new unmanned aircraft; RPAS two-way communications with ATC; operator able to contact ATC regarding special conditions; DSA; flight plan required.
VII	Very High Altitude IFR		VLOS and BVLOS	Above FL600, mostly uncontrolled airspace; transition FL600 and below through segregated or non-segregated airspace; pre-coordination; special procedures; flight plan required.

Under CAAC interim regulations, BVLOS is also allowed for Categories I and II UAS (under the new Category VII, which meets the FAA's sUAS weight limit), Category V agriculture UAS with an MTOW not exceeding 5,700 kg (12,500 lb) and operated within 15 m above the crop canopy, and Category VI airship with an envelope volume less than 4,600 m³.

Table 37: Civil Aviation Administration of China (CAAC) UAS Categorization.

Category	Empty Weight (kg)	Takeoff Weight (kg)	Operational Requirements		Remote Pilot
I	≤ 1.5		VLOS; CAS ≤ 100 km/h; < 120 m AGL; < 500 m from pilot/observer	Pilot assures safety.	No license required.
II	> 1.5, ≤ 4	> 1.5, ≤ 7		Geofencing and ground station Cloud connection with 60 s rate, if over or around important facilities or airports.	
III	> 4, ≤ 15	> 7, ≤ 25		Flight data stored for 3 months; geofencing; UAS Cloud connection with 1s (densely populated areas) or 30 s reporting rate.	Remote pilot license managed by a national professional organization. No license required for indoor operations. No license required for rural area experimental operations.
IV	> 15, ≤ 116	> 25, ≤ 150		Passive surveillance.	
V	Agriculture, W ≤ 5,700		BVLOS; ≤ 15 m	Geofencing and ground station Cloud connection with 60 s rate, if over or around important facilities or airports.	
VI	Airship, volume ≤ 4,600 m ³		BVLOS	Flight data stored for 3 months; geofencing; UAS Cloud connection with 1s rate (densely populated areas) or 30 s rate reporting.	
VII	BVLOS Categories I and II				

Note 1: For Categories I-IV, the category is determined by the higher requirements of the two.

Note 2: For formation flight, the category is determined by the total weight of all aircraft within the flight.

Note 3: For Categories I and II, use lower weight thresholds if so selected by local authority.

Note 4: For agriculture light and small UAS, altitude is limited to 15 m above crop canopy.

2.3.6 Fédération Aéronautique Internationale (FAI) UAV Categorization

For record flight purposes, the FAI has defined Class U Aircraft: Unmanned Aerial Vehicles (UAV) in 2001. The definition of Class U UAV is provided in Section 12 of FAI sporting code [49]. A UAV is an aircraft that does not carry a human and is controlled primarily by means of an onboard flight system. A UAV can be remotely controlled by a person or persons, either by direct sight or First Person View (FPV), or autonomously controlled by a hardware system and/or software system onboard the UAV, or both. Differences between FAI Aeromodelling and FAI UAV include: Model aircraft are always in VLOS of the pilot; For model aircraft, onboard flight systems that control position and attitude of the model aircraft are not permitted, except auxiliary stabilizing devices for certain classes.

Class U Aircraft is further divided into different classifications, as shown in Table 38. These classifications are defined to support distance, speed, duration, altitude, and payload record flights. As such, the focus of the classification are factors that directly impact those performance parameters.

Table 38: FAI Class U UAV Classification.

Classification	Code	Definition
Type (Subclass)	U-1	Type 1 Fixed wing aerodyne
	U-2	Type 2 Rotary wing aerodyne: Variable pitch (helicopter)
	U-3	Type 2 Rotary wing aerodyne: Fixed pitch (multirotor ≥ 3 rotors)
	U-4	Aerostat
	U-Absolute	Best records regardless of control, weight, and propulsion.
Weight (Category)	1	< 50 g
	2	≥ 50 g, < 250 g
	3	≥ 250 g, < 1 kg
	4	≥ 1 kg, < 2.5 kg
	5	≥ 2.5 kg, < 5 kg
	6	≥ 5 kg, < 25 kg
	7	≥ 25 kg, < 100 kg
	8	≥ 100 kg, < 500 kg
	9	≥ 500 kg, < 2,500 kg
	10	$\geq 2,500$ kg, < 10,000 kg
	11	$\geq 10,000$ kg, < 50,000 kg
	12	$\geq 50,000$ kg
Propulsion (Group)	Group 1	Internal combustion and Jet
	Group 2	Electric
	Group 3	Rocket
	Group 4	Unpropelled

2.3.7 Prior National Aeronautics and Space Administration (NASA) Research

One previous NASA study looked at the question of how to group UASs of similar physical, performance, or other characteristics, such that appropriate airworthiness standards, ultimately including reliability and design assurance requirements, can be assigned [64], [52], [65]. The study recognized the need to consider additional factors not considered in manned aircraft classification. A qualitative analysis was conducted to provide insights into the implications of different candidate UAS classification factors to safety risk, and potential use of these factors, as shown in [Table 39](#).

Table 39: NASA Summary of UAS Specific Classification Factors.

Item	Primary Aspect of Risk Affected by the Factor	Potential Benefits of Factor for Classification	Potential Issues with the Factor
Avionics Complexity	Likelihood	May account for increased dependence on safety-critical avionics	Clarity in defining complex
Airspeed	Severity	May account for expanded design space for UAS	Requires agreement in maximum airspeed definition
Kinetic Energy	Severity	May account for the amount of damage that can be inflicted upon people or other property on impact	Requires agreement in maximum speed definition
Operational Range (visual)	Likelihood	May account for new hazards related to using ground observers	Enforceability concerns; Clarity in defining loss of visual contact by occlusions
Operational Range (radio)	Likelihood	May account for new hazards from using radio control links	Enforceability concerns
Frangibility	Severity	May allow reduced accident severity through frangible structures that would be unacceptable for CPA	Clarity in defining frangibility
Operational Area	Likelihood	May account for hazards specific to airspace participants	Enforceability concerns
Overflown Area	Severity	May account for hazards to people and property on the ground	Enforceability concerns; Clarity in defining lightly inhabited areas
Operational Failure Consequence	Severity	May account for new hazards and risk assessment unique to UAS	Clarity in elaborating relevant failures
Altitude	Likelihood	May account for potential design differences for UAS propulsion or structural aspects relevant to flying at altitudes different from CPA	Enforceability concerns
Flight Time	Duration	May account for new hazards related to substantially different mission times (longer or shorter) for UAS	Requires agreement in definition of flight time
Aircraft Cost	Severity	May account for the quality of a UAS, in a way that is not necessary for CPA, since existing regulations for safety guarantee a level of quality	Clarity in defining aircraft cost; Unprecedented use of cost in 14CFR

In [Table 39](#), manned aircraft is referred to as *conventionally piloted aircraft* (CPA). The analysis considered four aspects relevant to risk: *likelihood*, the probability of a hazardous event; *severity*, the consequence of a safety event; *duration*, the length of time during which a hazardous event could occur; and *expectation*, relative level of acceptable risk. Factors relevant to the expectation were considered not UAS-specific. One of the conclusions from the wide range of potential benefits of those factors listed in the table is that different airworthiness standards may be needed to account for different design aspects. For example, a highly frangible UAS may have very different requirements with respect to structures and materials than one that is not intended to be frangible. In analyzing potential issues of the candidate factors, the team recognized that there was no objective definition for some of the issues at the time. It was also suggested that factors with issues should not be taken to mean that they are not usable. Instead, those issues should be considered if factors listed in [Table 39](#) are used in classification.

2.4 Lessons Learned from Existing Aircraft Categorization Methods

Through the comprehensive review of existing categorization methods and schemes for manned aircraft, model aircraft, and UAS, a few lessons could be learned:

- There is no single widely accepted existing categorization, either across different countries or regions, or across different organizations within the same country or region.
- There is no single existing categorization that serve all purposes. From a regulatory point of view, the main purpose is to manage safety risks, while from an application point of view, the main purpose is to manage mission and capabilities.
- No prior efforts have been found on UAS categorization in the context of UAS trajectory modeling and air traffic services.
- Manned aircraft categorization methods, although mostly a result of practical adaptation over time and not able to differentiate some aircraft configurations seen in sUASs, have also been tested over time. Additionally, through international harmonization efforts, such as those of the ICAO, common categorization methods have been more widely adopted by different countries and regions in recent years.
- The adaptation of manned aircraft classification, particularly that in airworthiness certification, demonstrated an upward movement of technical complexity over time, putting extra burden for aircraft towards the lower end of the spectrum for a given category. To cope with this issue, a category may be split to downgrade aircraft types in the lower end of the spectrum to a lower category; or a new, lower category may be introduced.
- New technology, lower thresholds to accessing more advanced technology, and/or market may also drive the introduction of new, more flexible categories to accommodate relatively new, simple, and less costly aircraft types that emerge and become popular.
- Model aircraft is the most relevant aircraft to sUAS in terms of size and performance. The FAI's well established model aircraft categorization, largely driven by international sports competitions through the charter of the FAI, is the most uniformly adopted categorization across different countries and regions. The specific airframe configuration, design, and safety requirements for different model aircraft categories provide a start point for determining sUAS categorization criteria.

- New technology and lower thresholds to accessing more advanced technology have demonstrated phenomena for model aircraft: 1) a new sUAS design may be first extensively tested and matured through model aircraft sports competitions; and 2) traditional categorizations or categories may become obsolete. For example, electric powered model aircraft used to have their own categories, but the boundary has become blurred as more traditional piston engine powered categories are now powered by electric motors with the same or better performance.
- Most diverse categorization methods have been observed for UASs, compared with those for model aircraft and manned aircraft. The military is focused on mission capabilities. Civil aviation regulatory authorities are more focused on safety risks. The technology, market, and regulation for UASs, particularly civil ones, are evolving at a fast pace; and the development is not uniform across different countries and regions. Nonetheless, existing UAS categorization methods used by various countries and regions, and various organizations, provide insights into the major factors to be considered.
- A recent trend is the proposed use of performance-based standards and a more comprehensive risk assessment to replace simple single factor (e.g., weight) or two-factor or three-factor classification schemes. Trajectory-based analysis is expected to play a more important role in risk assessment and UAS categorization.

In summary, a wealth of information and lessons can be learned from existing aircraft categorizations. However, there still lacks an sUAS categorization framework, particularly in support of air traffic services; and there is also a lack of data and analysis to validate the criteria used in some of the existing categorization schemes.

3 Small UAS Categorization Framework

With a comprehensive review of, and lessons learned from, various existing categorization methods and schemes for manned aircraft, model aircraft, and UASs, a starting point can be established for an sUAS categorization framework. Additional analyses of sUAS-specific characteristics to support the new sUAS categorization framework can also be identified. In this section, the requirements for an sUAS categorization framework in the context of trajectory management and traffic services is established first. The new sUAS categorization framework consists of categorizations from different aspects of the sUAS design and operations. Each of them will be discussed in a separate subsection.

3.1 Requirements for sUAS Categorization in the Context of Trajectory Modeling

Before the new sUAS categorization framework is discussed, it is necessary to determine the requirements for such a categorization framework, at least at a high level. In the context of sUAS trajectory management and traffic services, the categorization should recognize, differentiate, and understand various groups of sUASs from design, operational, and safety characteristics that influence significantly either the methods used in trajectory modeling, or the outcome of trajectory modeling, or both. Impact risks to other aircraft operating in the same airspace and risks to people and property on the ground have been the central concern for nearly all existing UAS categorization schemes. They are not directly considered in the new sUAS categorization framework presented here. It is assumed that the trajectory modeling capability being developed will support risk analysis and consequently influence sUAS categorization for regulatory and certification purposes.

Specifically, the sUAS categorization framework shall:

- Req-1** Recognize and differentiate aircraft configurations with unique flight dynamics
- Req-2** Recognize and differentiate power plants with unique performance characteristics
- Req-3** Recognize and differentiate the type of flights and consider any special handling of the trajectory in the UTM ecosystem
- Req-4** Recognize and differentiate flight rules for conducting sUAS operations
- Req-5** Recognize and differentiate flight operations with different levels of trajectory tracking accuracy
- Req-6** Consider the impact of different flight and operations control regimes
- Req-7** Consider the impact of different environmental conditions, nominal and off-nominal
- Req-8** Consider the impact of different system failures
- Req-9** Be flexible, extendable, and scalable, while affordable in support of sUAS trajectory management and traffic services

3.2 Aircraft Configuration Classification

A small UAS aircraft configuration classification is an adaptation of manned aircraft classification adopted by a regulatory authority such as the FAA. In the context of trajectory management and traffic services, the purpose of the sUAS aircraft configuration classification is to recognize and differentiate aircraft with unique flight dynamics and unique power plant performance characteristics ([Req-1](#) and [Req-2](#)). Due to the diverse design and configurations of sUASs, compared to manned aircraft classification, new aircraft configuration classes may have to be added. To be flexible, scalable, and affordable at the same time ([Req-9](#)), the classification must avoid enumerating aircraft configurations seen and foreseen in the sUAS arena. A criterion used in this process is that if the flight dynamics and performance characteristics for a specific aircraft configuration can be numerically derived from classes that are already in place, then a new class won't be introduced. The resulting sUAS aircraft configuration classification is shown in [Table 40](#).

Table 40: Small UAS Aircraft Configuration Classification.

Classification		Definition	
Lighter than Air	Airship		Engine-driven lighter-than-air aircraft that can be steered.
Heavier than Air	Fixed Wing	Glider	Lift generated by wing, but not depending principally on an engine for sustained flight, including powered gliders.
		Airplane	Lift generated by wing, engine-driven propulsion, including weight-shift control and powered parachute aircraft, regardless of launch and recovery methods.
	Rotor-craft	Helicopter	Lift and propulsion generated by engine-driven rotor(s), principally depending on cyclic pitch for pitch and roll control, including compound helicopters with forward flight thrusters.
		Multirotor	Lift and propulsion generated by engine-driven rotors, principally depending on differential lift from multiple rotors (normally fixed pitch) for pitch and roll control,
	Powered-lift		Capable of vertical takeoff, vertical landing, and low speed flight that depends principally on engine-driven lift devices or engine thrust for lift; and cruise flight that depends principally on wing for lift. May include gyrodynes.
	Other		Any other heavier than air aircraft configurations that may not fit or may not be derived from defined classes, for example ornithopters, or gyroplanes.

Compared to the manned aircraft classification shown in [Table 1](#) and [Table 2](#), the *balloon* is eliminated because it is already managed outside the regulation that governs sUASs even though a balloon might fall within the range of sUAS weight limits. The *kite* is eliminated for a similar reason.

The *airship* is retained because of its demonstrated capabilities and economic value, and the fact that it is already authorized for BVLOS operations in some countries, such as in China [\[16\]](#).

The *glider* is retained because it has always been part of the family of sUASs, and because of its unique flying procedures and trajectory characteristics.

The *weight-shift control* and the *powered parachute* are consolidated with the fixed wing *airplane*. Although their flight control mechanisms may be different from that of an airplane, for trajectory modeling purposes, they may be modeled as an airplane. Additionally, for sUASs, there is essentially no difference between flying a weight-shift control or powered parachute and an airplane, even if the aircraft is flown manually via remote control.

In addition to *helicopter*, *multirotor* is introduced as a new rotorcraft class. The main difference between these two is that a helicopter uses cyclic pitch as the main flight control mechanism for pitch and roll control, while a multirotor uses differential rotor thrust as the main flight control mechanism. Thus, the flight dynamics model and flight controls are different, and consequently have different implications on trajectory modeling.

The *powered-lift* is retained with an expanded scope to include both jet and rotor powered-lift systems. In practice, jet VTOL airplanes have been classified as airplanes for manned aircraft. The powered-lift class would represent the most diverse group of designs than any other explicitly defined classes. However, in general, it can be modeled by adding a transition mode between an airplane type cruise mode, and a helicopter or multirotor takeoff and landing mode.

The *gyroplane* and the *gyrodyne* are no longer considered separate classes. This is because the benefits offered by these two configurations over the *airplane* and the *helicopter* are very limited for the size of sUASs. Besides, the aerodynamic efficiency penalty incurred during cruise flight is significant. The *gyroplane* can either be modeled as an airplane with extraordinary stall characteristics, or it can be treated as a special class when it is necessary in some rare cases. The *gyrodyne* is essentially a *powered-lift* except that it becomes a *gyroplane* during cruise.

The differentiation among a landplane, a seaplane, and an amphibian can essentially be ignored for sUAS trajectory modeling (or for UASs in general). This is especially true when widely used sUAS launch and recovery mechanisms such as hand launch and catapult launch are considered. Takeoff and landing distances are generally very short for sUASs. As soon as the aircraft gains minimum sustainable speed, there will be no difference in trajectory, regardless of the launch and recover mechanisms.

Compared with other design characteristics, weight is less significant in how the sUAS trajectory may be modeled. This is due to the small range of MTOW an sUAS may have, compared with UASs in general and manned aircraft. Weight is still a very important factor in safety-risk-based sUAS categorization, particularly for operations over people. The intention here is to recognize the importance of the weight factor, but leave the specific considerations of the weight factor in sUAS characterization to those studies that focus on impact safety risks.

The four standard engine types for manned aircraft are adopted without any changes, as shown in [Table 41](#). All four engine types have been seen in sUAS applications, with the electric motor dominating the number of installations in recent years. For electric powered aircraft, the source of electricity may come from battery, fuel cell, solar cell, or hybrid systems with piston or turboshaft engine(s) and battery(s).

Table 41: Small UAS Engine Type Categorization.

Type Code	Engine Category	Propulsion
P	Piston Engine	Propeller, ducted fan, or rotor(s)
T	Turboprop/Turboshaft Engine	Propeller, ducted fan, or rotor(s)
J	Jet Engine	Jet (including bypass fan jet)
E	Electric Engine	Propeller, ducted fan, or rotor(s)

3.3 Type of Flight

For manned aircraft, simple and coarse categorization is used to group the type of flight into scheduled air service, non-scheduled air transport operation, general aviation, and military or others. A special handling code may be given as necessary to provide additional information about the type of flight, mostly for general aviation and military flights.

For sUASs, scheduled air services probably won't be as dominant as those for manned aircraft flights. Most sUAS flights, including non-scheduled air transport operations such as package delivery, may correspond to the general aviation flight category. As such, the existing type of flight categorization for manned aircraft operations has very limited utility for sUAS operations. The diverse types of flights to be carried out by sUASs, the diverse operating environment for different sUAS applications, and their implications in sUAS trajectory require a different approach for categorizing the type of flight.

The proposed categorization is based on the area where the sUAS flight is carried out, as shown in [Table 42](#). The operation areas as defined in the table have strong implications for the nominal or off-nominal trajectories. Under the same weather condition, each of the areas will impose unique environmental conditions to sUAS operations, such as localized wind field, temperature, pressure, and obstructions by manmade structures. The *urban area* is defined as the airspace volume near the boundary of or below the urban canopy. An sUAS flight well above the urban canopy is considered a populated area operation. Flight over open-air assembly of people is considered a separate category regardless of the geographic area, whether rural, populated, or urban. For this category, trajectory assessment must be performed for all possible scenarios and situations. This categorization offers flexibility to accommodate operational categories such as those proposed by the FAA [\[30\]](#) ([Table 27](#)) and the EASA [\[22\]](#)([Table 35](#)).

Analogous to the special handling information for manned aircraft flight, a mission objective categorization is also proposed to provide supplemental information about the flights, as shown in [Table 43](#). This information may be valuable in aiding situation awareness for stakeholders that have an interest in the area where the sUAS flight is carried out. For example, a highway patrolling sUAS flight is of little concern to a crop dusting sUAS flight if the working field for the latter is well separated from the highway, while a package delivery sUAS flight may be a concern to the crop dusting sUAS flight if the package delivery destination is on the other side of the crop dusting working field. Although several scenarios have already been considered [\[72\]](#), additional analysis is needed to complete the list of proposed categories for the mission objective supplemental information.

Table 42: Small UAS Type of Flight.

Code		Category	Description
Type of Flight	R	Rural Area Operation	Open airspace, minimum probability of hitting people or property on ground in case of a complete system failure.
	P	Populated Area Operation	Open airspace with moderate obstructions of manmade structures, medium probability of hitting people or property on ground in case of a complete system failure.
	U	Urban Area or Dense Industrial Complex Operation	Near the boundary of or below the urban canopy with dense obstructions of manmade structures, high probability of hitting people or property on ground in case of a complete system failure.
	O	Over Open-air Assembly of People Operation	Directly over assembly of people in a laterally confined open area, high probability of hitting people or property on ground, possibility of secondary impact.

Table 43: Small UAS Mission Objective Supplement Information.

Code		Category	Note
Mission Objective Supplement	HUM	Humanitarian mission	To provide concise information aiding stakeholder situation awareness. Specific categories will be determined pending additional analysis.
	MED	Life critical medical delivery	
	NSP	Infrastructure inspection	
	RPT	News and media report	
	SAR	Search and rescue mission	
	STA	Military, customs or police services	
	SWM	Formation or swarming flight	

3.4 Flight Rules

For manned aircraft flights, there are two sets of flight rules, IFR and VFR. The distinction between the two is based on the pilot's use of outside visual reference. This distinction faces challenges for UAS flight operations because the pilot is not onboard the aircraft. For UASs, meteorological conditions are not the sole factor in determining the remote pilot's visual reference. Daylight-only operations are frequently specified in current UAS regulations. Sometimes a distance limit such as 500 m (1640 ft) from the remote pilot is specified, as in the UK [13] and China [16]. Although flight rules for sUAS VLOS operations are well established, such as those by the FAA [27], flight rules for BVLOS operations, are not yet well established.

The EVLOS operations as defined by the UK CAA [13], allow UAS operations either within or beyond 500 m lateral distance (the VLOS distance limit) from the remote pilot, using alternative means to maintain direct visual contact with the unmanned aircraft. The remote pilot is still responsible for visually complying with collision avoidance, and maintaining positive control at all times. However, there are no established standards or rules on the alternative methods and procedures. It is up to the operator to submit a safety case, including a risk assessment that addresses the procedures for avoiding collisions, aircraft aids to observation, and the use of deployed observers, among other factors, before a permission is considered. If a visual observer is required for EVLOS, then it is still within the FAA’s definition of VLOS.

Although the FPV technology is not considered a means to meet current VLOS requirements, comments received in response to the FAA’s notice of sUAS rulemaking argued the capability, field test, and the rapid development of the technology, as well as some of the limitations of FPV compared with direct human visual contact [27]. Recent developments, particularly those driven by public and commercial FPV drone racing events, have significantly advanced the FPV technology in terms of capability and reliability, including in BLOS communication environments, although not necessarily for long distance use. Although some limitations would still exist in the foreseen future, the FPV technology is arguably significantly more mature and less complex than sense-and-avoid (SAA) technologies that target full UAS integration into the NAS. It is thus valid to consider using FPV as an intermediate step to full BVLOS operations.

The proposed flight rules for sUASs include three categories as listed in Table 44. VLOS flight rules are considered the same as in Part 107. The FPV BVLOS, still conducted under visual meteorological conditions, is a set of flight rules bridging VLOS and full BVLOS that may be conducted under instrument meteorological conditions. It is envisioned that such a division will provide three unique sets of conditions for trajectory modeling.

Table 44: Small UAS Flight Rules.

Category	Meteorological Condition	Description
VLOS	Visual	Rely on remote pilot maintaining direct visual contact (including via an observer) for collision avoidance and contingency.
FPV BVLOS	Visual	Rely on remote pilot utilizing FPV video feed, instrument telemetry, and potentially enhanced vision for collision avoidance and contingency; pilot positive control maintained at all times; geofencing may be required.
Instrument BVLOS	Visual or Instrument	Rely on instrument telemetry, potentially synthetic vision, and onboard autonomy for collision avoidance and contingency; geofencing required.

For the FPV BLOS, the FPV camera may be mounted on pan-tilt gimbals to enhance the scan of the airspace. Instrument telemetry is required under this set of flight rules to provide redundancy and real time system health information to the remote pilot. Instead of requiring full onboard autonomy, the remote pilot is required to maintain positive control of the aircraft at all times. The FPV and instrument telemetry may need to be subject to a regulatory or community consensus-based airworthiness certification process. A flight plan approval process may also need

to be implemented to address potential communication link conditions in certain airspace areas. The onboard electronic target recognition and processing are not mandated for FPV BLOS.

To differentiate it from FPV BVLOS, the future full BVLOS is called Instrument BVLOS. FPV video feed may still be used by the operator, but it is not required, nor is it considered a means to meet sense-and-avoid requirements. Onboard electronic target recognition and processing are part of the sense-and-avoid requirements. Increased level of onboard autonomy may be required to address collision avoidance and contingency.

3.5 Equipage and Capabilities

Because of the diverse radiocommunication and navigation technologies and systems being developed and manufactured for sUAS applications, unlike that for manned aircraft, there is no standard equipage, installation, protocols, or approval standards exist. The current trend is to utilizing performance-based and risk-based standards, instead of prescriptive standards to mandate specific equipage, installation, or protocols. As such there is not yet a strong case to propose radiocommunication and navigation categorization for sUAS.

On the other hand, to address the security and DAA concerns for sUAS operations such as operation over people, night operations, and BVLOS operations, there is a real need to establish and identify remote identification (ID) and tracking [11], [47], and surveillance [38] equipage and capabilities. To ensure inter-operability between sUAS vehicles, UAS service suppliers (USSs), and air navigation service providers (ANSPs), standardized equipage and installation may need to be established. The proposed sUAS equipage and capabilities are shown in [Table 45](#).

The surveillance systems 1,090 MHz ADS-B, UAS ADS-B, and local remote ID operate in a similar manner. That is, the broadcast may be received by any equipped systems, airborne or ground. The network remote ID operates over a point to point connection between the sUAS and the USS network remote ID service. The connection between the sUAS vehicle and the USS network remote ID service, if it exists, may be a direct wireless network connection or may be achieved through the sUAS ground control station (GCS).

3.6 Performance-Based Navigation (PBN) Capabilities

For manned aircraft operations, the PBN concept specifies that aircraft RNAV or RNP system performance requirements shall be defined in terms of the accuracy, integrity, continuity, and functionality required for the proposed operations in the context of a particular airspace concept. The PBN concept can be easily applied to UASs that are intended to be operated in manned and unmanned mixed airspace environment, but it does not scale well for sUAS operations. On one hand, even the standard approach RNP 0.3 (1823 ft/556 m) and minimum RNP 0.1 (607 ft/185 m) are too large for sUASs considering their size and their intended applications. On the other hand, an sUAS is subject to the same navigation system error (NSE) as manned aircraft, if not worse. There have been discussions about PBN (or RNP in most cases) for sUASs at various occasions. In a recent NASA study [60], a quantitative framework to correlate operational risk with trajectory conformance requirements was developed. Additional research is needed before a standard is defined for sUAS operations. It would not be a simple undertaking.

Table 45: Small UAS Equipage and Capabilities.

Code		Description	Notes
Surveillance	N	No surveillance equipment	
	B1	1,090 MHz ADS-B Out	Standard 1,090 MHz ADS-B equipage and capability.
	B2	1,090 MHz ADS-B Out and In	
	B3	1,090 MHz ADS-B In	
	U1	UAT ADS-B Out	Standard UAT ADS-B equipage and capability.
	U2	UAT ADS-B Out and In	
	U2	UAT ADS-B In	
	L1	Local Remote ID Out	Local broadcast remote ID and tracking capability, similar to ADS-B, but specifically established for sUAS operations below 400 ft AGL.
	L2	Local Remote ID Out and In	
	L3	Local Remote ID In	
	N1	Network Remote ID Out	Remote ID and tracking capability available over the network that is either via direct sUAS vehicle connection or via the sUAS GCS.
	N2	Network Remote ID Out and In	
	N3	Network Remote ID In	
	R	Non-cooperative sensor(s)	
DAA	D0	No DAA capability	No automated DAA, including VLOS operations.
	D1	Airborne DAA	DAA hosted by an airborne system.
	D2	Ground-based DAA	DAA hosted by a ground-based system.

Within this research, an attempt is carried out to define a framework for categorizing the TSE, or the accuracy aspect of sUAS PBN capabilities, all assuming the same basic PBN concept for manned aircraft. The complete RNP requirements should be defined in terms of accuracy, integrity, continuity and functionality, including aircraft and pilot as a system, with significant adaptations to the specific characteristics of sUAS operations. For this reason, only the general term *PBN* is used for sUASs. The term *RNP* is not used in this discussion.

The proposed sUAS horizontal PBN capability categorization framework is shown in [Table 46](#). The horizontal PBN specification is the accuracy in terms of TSE in meters in the lateral or longitudinal direction, which is expected to be achieved at least 95% of the flight time. The accuracy is measured from a desired path, if such a path is prescribed for the flight. Or it may be measured from a dynamically defined flight path, such as when the sUAS is conducting an infrastructure inspection flight. In the latter case, the flight path may be defined by a path that maintains an ideal distance from the infrastructure, which may not have a prescribed geometry that is known or accessible to the sUAS prior to the mission. Another example is obstacle avoidance

against stationary objects. The horizontal PBN descriptors are chosen such to avoid confusion with existing ICAO PBN descriptors. Three descriptors are proposed to represent three levels of PBN specification. Additional levels may be added if justified by analysis.

The PBN concept includes specification of the NAVAID required to meet the performance. For sUASs, Global Positioning System (GPS) and inertial measurement unit (IMU) have been the de facto standard systems. Traditional radio navigation is not used, presumably due to their low accuracy and signal reception at low altitudes. Various localized navigation systems employing radio frequency, laser, or vision sensors have been developed either as a complement to GPS or as a substitute in GPS challenged/denied environments. Miniaturized relative navigation systems have also been used by sUASs, often as part of an obstruction collision avoidance system. These two navigation systems will likely be incorporated in to the sUAS PBN specification when it is fully developed. Similar to the ICAO PBN specification, index numbers may be attached to each descriptor shown in [Table 46](#) to indicate the required NAVAID or specific procedure.

Table 46: Small UAS Horizontal PBN Capability.

Descriptor	PBN Specification	Description
X	60 m (200 ft)	Preliminary baseline PBN specification for sUAS operations.
Y	30 m (100 ft)	Tentatively given as 50 % of the baseline PBN specification, may be applied to sUASs en route or mission operations.
Z	Standard: 15 m (50 ft), Precision: 3 m (10 ft)	Minimum PBN specification, for precision mission tasks, departure and approach operations. The standard minimum is for general applications, and the precision minimum is for precision departure, approach, hovering, or other relative navigation-based mission tasks.

The lowest PBN specification for manned aircraft operations is RNAV 10 (RNP 10). RNAV 10 is designed for oceanic and remote operations and it does not require onboard performance monitoring and alerting. Thus, RNAV 10 can be considered the baseline PBN specification for manned aircraft operations. A challenge exists for defining the baseline sUAS horizontal PBN specification because no existing sUAS separation minima have been defined yet. It is an active topic of research.

A recent study [78] recommended that the *well clear* for sUASs vs. manned aircraft be defined as a horizontal miss distance (HMD) of 2000 ft (610 m) and vertical miss distance (VMD) of 250 ft (76 m). It is also stated that the HMD is not sensitive to the sUAS trajectory. Compared with RNP 0.3 (1822 ft, 556 m) and RNP 0.1 (608 ft, 185 m) for manned aircraft, the HMD of 2000 ft has little reference value to defining a baseline sUAS horizontal PBN specification.

On the other hand, through years of continued practice and assessment, the FAI has defined minimum distances between the safety line and pilot, judges, or spectators for various RC model aircraft classes [48]. For F3A standard (5 kg), F3M large (25 kg), and F3S jet (5 kg) aerobatic aircraft, the safety line (no-fly zone) is located at 30 m (98 ft) in front of the pilot, while judges shall be seated 7-10 m behind the pilot. For the F3N freestyle helicopter (6.5 kg), the prohibited flying area extends 15 m from the judges, and 25 m from spectators. The safety line distances are smaller than those for airplanes.

The FAI F3D pylon racing (3 kg without fuel) is the most advanced FAI model aircraft class, with an average course speed of up to 65 m/s (234 km/h, or 145 mph). The sideline, outside which it is considered safe, is 90 m from pylon 2 or 3, the closest pylon of a three-pylon course. Because model aircraft fly outside the triangle defined by the three pylons, the desired flight path will not be exactly flying over the pylons. There is a normal track dispersion even when the competitor tries to fly a track as tight as appropriate to make a 10-lap race in the shortest amount of time. Obviously, from continued assessment, the FAI believes that a lateral distance of 90 m is sufficient to contain the visual, manually flown track with a sufficient safety buffer. In addition to the FAI's safety rules for standard aircraft classes, the AMA has defined safety rules for its largest national model aircraft classes, i.e., LMA-1 (55-77.2 lb) and LMA-2 (77.3-125 lb) large model airplanes and LTMA-1 (55-77.2 lb) and LTMA-1 (77.3-100 lb) large turbine model airplanes [6]. The AMA requires a minimum distance of 50 ft (15 m) between the takeoff and landing path and the pilot line, and a minimum distance of 200 ft (60 m) between all other flight paths and spectators. Note that normal track dispersion does not have to be considered for the AMA large airplane classes. With that in mind, the AMA safety rules are generally in line with FAI safety rules.

The FAI has also established landing requirements for various RC model classes [48]. For the F3B multi-task glider, points will be awarded for landing based on the distance from the marked landing spot and the glider. One hundred points will be awarded if the glider lands within 1 m from the center of the marked landing spot, fewer points will be awarded as the distance increases until the distance becomes more than 15 m, when 0 points will be awarded. The F3J thermal glider has a similar landing award system except that it emphasizes more precision landings than the F3B. Note that the distance from the landing spot reflects a combination of both lateral and longitudinal errors.

For the FAI F3C helicopter, points will be awarded for landing based on circles marked on the helipad, 1 m and 3 m in diameter, respectively. For powered landing, a maximum score will be given if the landing skid is completely within the 1 m circle. Two points will be downgraded if the main rotor shaft is outside the 1 m circle. For autorotation landing, if the rotor shaft is inside the 1 (or 3) m diameter circle, 4 (or 2) points will be given. If the rotor shaft is outside the 3 m circle, 0 points will be given. Related, for the F3U FPV racing, the air gates that the quadcopter must fly through have a maximum width of 3 m and minimum width between 2.5 m and 1.6 m. These rules represent the highest lateral control precision required for a model aircraft.

Based on the discussion about model aircraft landing precision rules, 50 ft (15 m) is selected as the preliminary standard minimum horizontal PBN specification for sUASs, and 10 ft (3 m) is selected as the preliminary precision minimum horizontal PBN specification. As indicated in Table 46, the standard minimum may be used for general applications, such as in pipeline inspection. The precision minimum may be used for departure, arrival, hovering, or other relative navigation-based mission tasks, such as package drop off, industrial inspection, etc. Note that for manned helicopters, to simplify the process, a single RNP 0.3 is proposed for all phases of flights including terminal and en route operations to benefit the majority of the current fleet.

For the intermediate level of horizontal PBN specification, 100 ft (30 m) is tentatively selected, which is half of the baseline horizontal PBN specification, and about 10 times the precision minimum horizontal specification.

To fully develop an sUAS horizontal PBN specification, additional analyses are required. Firstly, operational requirements must be well understood. Such understanding may be developed

by a survey of the needs of various sUAS applications and missions, and an analysis of airspace and traffic management operational scenarios. The latter may include lateral obstacle clearance, separation minima, route spacing, and traffic flow management requirements. Secondly, what could be reasonably achieved needs to be carefully assessed. This could follow practices employed for manned flight operations; simulation studies with various navigation, guidance, and control capabilities; and vehicle performance similar to those recommended in [60]. The challenge is a lack of a sufficiently large volume of operational data. Trajectory modeling capability becomes an important foundation for this analysis.

So far the sUAS PBN categorization has been limited to the lateral and longitudinal directions, or in the horizontal plane. The word *horizontal* is deliberately used in the discussions above to reflect this nature. However, for low altitude sUAS operations, the vertical PBN may be equally important, if not more. Confined within 400 ft (122 m) AGL, the altimeter setting, the equipage of radio or lidar altimeter, and the accuracy of GPS altitude represent a few challenges faced by sUAS operations, particularly Instrument BVLOS. For manned flight operations, vertical RNP is neither defined nor included in the PBN concept. For sUASs, separation minima, including the ones in the vertical direction, are not yet defined either. Additional analysis is required to define a best path for sUAS vertical navigation performance. This could potentially include vertical PBN specifications, performance-based vertical separation minima (VSM) analogous to the reduced vertical separation minimum (RVSM) approval process, or community consensus-based airspace classification below 500 ft (152 m). A notional classification is provided in [Table 47](#) as a placeholder.

Table 47: Small UAS Vertical PBN Capability.

Category	Vertical Separation Minimum	Note
Standard	30 m (100 ft)	E.g., utilizing a combination of GPS altitude and barometer.
Precision	15 m (50 ft)	E.g., with additional en route altimeter setting or localized navigation services. May also employ sUAS-sUAS SAA.

With the PBN capability categorization proposed above, one may avoid having to include specific navigation and control equipage information. This is particularly advantageous, because unlike manned aircraft, great variation in terms of performance exists for sUASs.

3.7 Flight and Operations Control Categorization

Much like manned aircraft, an sUAS flight may be carried out by the remote pilot completely manually, or utilizing different levels of automation. Additionally, an sUAS flight may be carried out autonomously, including generating flight path or flight intent during flight without human pilot intervention. For large manned commercial operators, there is a distinction between flight operations (pilot) and operations control (dispatcher or operations control specialist), and there is a joint flight safety responsibility. For sUASs (and light UASs as well), flight operations and operations control may be combined and may be difficult to distinguish from each other. As such, a single categorization is proposed for sUAS flight operations and operations control, as shown in [Table 48](#).

Categories in [Table 48](#) are listed in the order of increased levels of system complexity, automation, and autonomy capability. It is possible that a particular sUAS may allow mixed use of

different categories in a given flight. As such, the categorization may be used to categorize the capability of an sUAS, a flight carried out by an sUAS, or different phases of a flight, in a way similar to the categorization of PBN capabilities. In general, it is expected that a more capable sUAS may be capable of all the less complex control categories. That is, an Autonomous Control (AC) capable sUAS may be operated in Flight Plan Control (PC), Mode Control (MC), and Direct Control (DC) modes. But this may not always be true, depending on the specific system design.

It should be noted that the flight and operations control categorization is not tied to the flight rules. For example, VLOS operations may be carried out in AC, PC, MC, or DC, as long as the remote pilot maintains positive control of the sUAS at all the times. This means that, if a flight is being conducted in AC, the pilot shall be able to switch to PC, MC, or DC at any moment during normal flight. Theoretically, BVLOS may also be carried out in DC with the remote pilot manually flying the sUAS during the entire flight.

Table 48: Small UAS Flight and Operations Control Categorization.

Category	Description	Note	
Direct Control (DC)	A broad category within which the remote pilot directly manipulates flight control surfaces via RC link to achieve desired attitude, speed, altitude, and lateral path. Electronic stability augmentation systems may be utilized to reduce pilot workload and enhance safety.	High pilot action variability impact on trajectory.	Ground control owns intended trajectory
Mode Control (MC)	The remote pilot provides high level control commands via RC link, such as turn rate (bank angle), climb rate, target altitude, etc., while the manipulation of control surfaces being carried out by autopilot functions.	Reduced pilot action variability impact on trajectory.	
Flight Plan Control (PC)	The remote pilot provides an initial flight plan, and subsequent flight plan amendments during the flight via data link, while the manipulation of control surfaces being carried out by autopilot functions following high level control commands generated by guidance functions.	Eliminated pilot action variability impact on trajectory.	
Autonomous Control (AC)	The remote pilot provides an initial flight plan and/or preprogramed or event driven uplink command to engage autonomous flight control logic that will generate flight intent or flight plan modifications based on sensor data to automatically carry out the flight utilizing guidance and autopilot functions.	Significant situation variability impact on trajectory.	Airborne system (partially) owns intended trajectory

3.8 Flight Performance Categorization

The FAA's sUAS final rule, 14 CFR Part 107, imposes an 87 kt (100 mph) limit on the groundspeed, rather than on the airspeed. This speed limit is determined based on the potential risk to person, property, and other aircraft in case of loss of positive control [27]. In justifying the speed limit, it was argued that a speed limit of 87 kt (100 mph) must be viewed within the context of the

overall regulatory framework of Part 107. The remote pilot in command must, prior to flight, assess the operating environment and consider risks to persons and property in the vicinity both on the surface and in the air. This includes, for an operation with a planned maximum speed, as part of the preflight assessment process, the pilot taking precautions to ensure that the sUAS will remain within visual line of sight, and that it will not pose an undue hazard to other aircraft, people, or property on the ground. Those precautions will likely be greater for an operation with a higher planned maximum speed. Additionally, within the imposed speed limit, winds and turbulence will have a disproportionately higher impact on sUASs than they would on larger manned aircraft flying at higher speeds. For the same airspeed, a strong tail wind will significantly increase the groundspeed, causing the sUAS to exceed the speed limit if it is flying fast enough. Conversely, a strong head wind will significantly decrease the groundspeed, causing the sUAS to lose groundspeed, and thus having difficulties returning to its base. Based on these discussions, it is thus necessary to establish a maximum speed categorization.

The proposed sUAS speed categorization is shown in [Table 49](#). This categorization is based on maximum CAS. For a powered sUAS, this is the maximum CAS that can be achieved during level flight. For an sUAS glider, this is the maximum never-exceed speed. As an initial proposal, four categories are defined. Each of the categories is expected to cover a reasonably large portion of operational regimes or a reasonably large portion of the sUAS fleet. The proposed category code consists of the letter A, which is the same as the ICAO aircraft category code for approach speed less than 91 kt (105 mph), and a numeric digit that differentiates the sUAS speed category.

Table 49: Small UAS Speed Categorization.

Category	Maximum CAS	Explanation
A1	< 22 kt (25 mph)	Intended to cover the slow flying sUAS fleet. Speed criterion to be determined by the ability to maintain safe flight in nominal wind and gust conditions.
A2	≥ 22 kt (25 mph), < 43 kt (50 mph)	Intended to cover the intermediate performance sUAS fleet, which are expected to comprise the largest portion of the entire sUAS fleet.
A3	≥ 43 kt (50 mph), < 65 kt (75 mph)	Intended to cover the more advanced performance sUAS fleet, which are expected to comprise a substantial portion of the BVLOS operations sUAS fleet and of the entire sUAS fleet as well.
A4	> 65 kt (75 mph)	Intended to cover the high performance sUAS fleet. Speed criterion to be determined by the requirement to conduct normal flight operations without exceeding the 87 kt (100 mph) groundspeed limit under normal tail wind conditions.

[Table 49](#) also provides explanations and criteria for each category. The A1 category is intended to cover the slow flying sUAS fleet. This category may include a large portion of the lower-end sUAS fleet, both fixed wing and rotorcraft, that are intended for VLOS operations. An sUAS in this category may be suitable for operations under nominal wind and gust conditions only, which is similar to the conditions under which most hobbyists fly their RC model aircraft.

The A4 category is intended to cover the high performance sUAS fleet. These aircraft are most likely intended for BVLOS operations. Operators of sUASs in this category must carefully assess the operating environment and consider risks prior to each flight. Particularly, the remote pilot

should take precautions to ensure that the sUAS will remain within in the 87 kt (100 mph) groundspeed limit during normal flight operations, including under normal tail wind conditions. Due to atmospheric boundary layer effects, the wind speed at the cruise altitude, e.g., 350 ft, could be twice the value, or more, of the wind speed indicated in weather forecasts and weather reports, which are normally the wind speed at 10 m (typically 33 ft or 27 ft in the U.S.) AGL.

The other two categories in between are intended to cover the intermediate performance sUAS fleet and the more advanced performance sUAS fleet, respectively. It is expected that these two categories will cover the majority of the overall commercial sUAS fleet.

The proposed categorization shown in [Table 49](#) is based on the preliminary analysis discussed above. The granularity and the speed criteria of the fully developed sUAS speed categorization should be refined based on additional in-depth analyses. A survey of the sUAS products available in the market may be conducted to compare the maximum speed and the max wind speed under which the sUAS may be operated. For example, in sports mode, the DJI Phantom 4 quadcopter has a maximum speed of 20 m/s (39 kt/45 mph) and a maximum wind resistance speed of 10 m/s (19 kt/22 mph). A regression and clustering analysis may provide insights into the criteria used in the categorization. Trajectory simulation analysis may also be conducted to provide quantitative results to support the justification. It is worthy to point out that the performance of flight controls is an important factor in determining the sUAS wind resistance capability. The simulation analysis should be based on a nominal level flight control system to avoid the results being biased as overly optimistic.

In addition to horizontal winds, sUASs are also subject to up draft and down draft due to terrain, obstructions of manmade structures, trees, thermals, and breeze fronts etc. Additionally, vertical maneuvers may be required to yield the right of way in traffic situations. The maximum vertical rate is an important factor in maintaining or changing altitudes during flight. A proposed preliminary vertical rate categorization is shown in [Table 50](#).

Table 50: Small UAS Vertical Rate Categorization.

Category	Maximum Vertical Rate	Note
N	±3 m/s (600 fpm)	Normal, nominal vertical maneuverability.
U	±7 m/s (1400 fpm)	Utility, intermediate vertical maneuverability.
H	±10 m/s (2,000 fpm)	High performance, agile vertical maneuverability.

Other flight performance factors to consider may include WTC. Additional analysis is required to consider lateral and vertical separation minima, wake turbulence, formation or swarming flying, ground effects, and effects of nearby structures in a holistic manner.

4 Summary

Through this effort, categorization methods applied to manned aircraft and existing UAS categorization methods (including model aircraft) were examined to provide a starting point for additional analysis of sUAS-specific characteristics. Based on that analysis, particularly implications of various factors on sUAS trajectories, a sUAS categorization framework was established to support trajectory management and traffic services. Unlike many prior UAS categorization studies geared towards airworthiness certification or operational approval that resulted in a single categorization in each case, the sUAS categorization framework proposed in this research includes categorization methods for each of the following aspects:

- Aircraft configuration
- Type of flight
- Flight rules
- Equipage and capabilities
- Performance-based navigation (PBN)
- Flight and operations control, and
- Flight performance.

Initial criteria for the categorization methods were selected based on either the limited data from the review of existing manned and unmanned aircraft categorization methods, or on a preliminary quantitative or qualitative analysis. The categorization framework is expected to be iteratively improved and refined as the understanding of sUAS-specific operational characteristics deepens and the trajectory modeling research progresses. Numerical analysis required to support the determination of various categorization criteria are identified and suggested for the final determination and incorporation of the categorization framework into UTM operations.

Such a categorization framework allows for sUAS operations to be managed efficiently, considering operating conditions and efficiency and safety objectives, based on distinctive sUAS characteristics recognized according to the established framework.

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Acronyms and Abbreviations

0

2D Two-dimensional

A

AC Autonomous Control
 ACARS Aircraft communications addressing and reporting system
 ACAS Airborne collision avoidance system
 ACAS X ACAS as a successor of TCAS
 ACAS sXu ACAS for sUAS
 ACAS Xu ACAS for UAS
 ACID Aircraft identification
 ADF Automatic direction finder
 ADG Airplane design group
 ADS-B Automatic dependent surveillance broadcast
 ADS-C Automatic dependent surveillance contract
 AGL Above ground level
 AIS Abbreviated Injury Scale
 AMA Academy of Model Aeronautics
 ANO Air navigation order (CAA)
 ANSP Air navigation service provider
 APCH Approach
 APV Approach with vertical guidance
 AR Authorization required
 ARC Aviation rulemaking committee
 ATC Air traffic control
 ATCT Air traffic control tower
 ATFM Air traffic flow management
 ATN Aeronautical telecommunication network
 ATS Air traffic service

B

BALL Balloon (ICAO designator)
 BARO-VNAV Barometric vertical navigation

BDS BeiDou Navigation Satellite System
 BLOS Beyond line of sight
 BVLOS Beyond the visual line of sight

C

CA Collision avoidance
 CAA Civil Aviation Authority (UK)
 CAAC Civil Aviation Administration of China
 CAS Calibrated airspeed
 CBO Community-based organization
 CFR Code of Federal Regulations
 ConOps Concept of operations
 CPA Conventionally piloted aircraft
 CPDLC Controller pilot data link communications

D

DAA Detect and avoid
 D-FIS Digital flight information service
 DC Direct Control
 DME Distance measuring equipment
 DoD Department of Defense
 DSA Detect, sense, and avoid

E

EASA European Aviation Safety Agency
 EFB Electronic Flight Bag
 EUROCONTROL European Organisation for the Safety of Air Navigation
 EVLOS Extended visual line of sight (CAA)

F

FAA Federal Aviation Administration

FAI	Fédération Aéronautique Internationale	L	
FANS	Future air navigation system	LMA	largest model airplane
FL	Flight level	LORAN	Long range air navigation system
FMC	Flight management computer	LORANC	LORAN C
FPV	First person view	LOS	Line of sight
		LPV	Localizer performance with vertical guidance
G		LSA	Light-sport aircraft
GBAS	Ground-based augmentation system	LTMA	Large turbine model airplane
GCS	Ground control station	M	
GLID	Glider (ICAO designator)	MAA	Military Aviation Authority (UK)
GNSS	Global navigation satellite system	MALE	Medium altitude, long endurance
GPS	Global Positioning System	MC	Mode Control
GYRO	Micro-/ultra-light Gyrocopter (ICAO designator)	MLS	Microwave landing system
		M _{MO}	Maximum operating limit Mach
H		MNPS	Minimum navigation performance specification
HALE	High altitude long endurance	MOC	Means of compliance
HF	High frequency	MOD	Ministry of Defence (UK)
HFDL	HF data link	MTOM	Maximum takeoff mass
HMD	Horizontal miss distance	MTOW	maximum takeoff weight
HTOL	Horizontal take-off and landing	MTSAT	Multifunctional Transport Satellites
		μUAS	Micro small UAS
I		N	
IAS	Indicated airspeed	NAA	National Aeronautic Association
ICAO	International Civil Aviation Organization	NAA	National aviation authority
ID	Identification	NAC	National Airport Control
IFR	Instrument flight rules	NAS	National Airspace System
ILS	Instrument landing system	NASA	National Aeronautics and Space Administration
IMU	Inertial measurement unit	NATO	North Atlantic Treaty Organization
INS	Inertial navigation system	NAVAID	Navigation aid
IRS	Inertial reference system	NSE	Navigation system error
IRU	Inertial reference unit	O	
J		OCC	Operations Control Center
JARUS	Joint Authorities for Rulemaking of Unmanned Systems	OSC	Operating Safety Case (CAA)
		OSD	Operational services and environmental definition
K			

P

PBN	Performance based navigation
PC	Flight Plan Control
PDC	Pre-departure clearances
PIC	Pilot in command

Q**R**

RC	Radio control
RCP	Required communication performance
RECAT	Wake turbulence recategorization
RF	Constant radius arc to a fix
RNAV	Area navigation
RNP	Required navigation performance
RPA	Remotely Piloted Aircraft (USAF)
RPAS	Remotely piloted aircraft system
RTCA	Radio Technical Commission for Aeronautics
RTF	Radiotelephony
RVSM	Reduced vertical separation minimum
RWC	Remain well clear

S

SAA	Sense-and-avoid
SATCOM	Satellite communication
SATVOICE	Satellite voice
SBAS	Satellite-based augmentation system
SHIP	Airship (ICAO designator)
SUA	Small unmanned aircraft (CAA)
sUAS	Small Unmanned Aircraft System
SUAS	Small UAS (USAF)
SUSA	Small unmanned surveillance aircraft (CAA)

T

TACAN	Tactical air navigational aid
TBD	To be determined

TC

TCAS Traffic alert and collision avoidance system

TDG

Taxiway design group

TSE

Total system error

U

UAS Unmanned aircraft system

UAT Universal access transceiver

UAUH Micro-/ultra-light Autonomous Unmanned Helicopter (FAA designator)

UAUL Micro-/ultra-light Autonomous Unmanned Aircraft (FAA designator)

UAV Unmanned aerial vehicle

UHEL Micro-/ultra-light Helicopter (ICAO designator)

UHF Ultrahigh frequency

UK United Kingdom

ULAC Micro-/ultra-light aircraft (ICAO designator)

UOPH Micro-/ultra-light Optionally Piloted Helicopter (FAA designator)

UOPL Micro-/ultra-light Optionally Piloted Aircraft (FAA designator)

URPH Micro-/ultra-light Remotely Piloted Helicopter (FAA designator)

URPL Micro-/ultra-light Remotely Piloted Aircraft (FAA designator)

USAF U.S. Air Force

UTM UAS traffic management

V

VDL VHF data link

VFR Visual flight rules

VHF Very high frequency

VLA Very light airplane

VLOS Visual line-of-sight

VMD Vertical miss distance

V_{MO} Maximum operating limit speed

V_{NO} Maximum structural cruising speed

VOR VHF omnidirectional range

VSM	Vertical separation minima	X	
VTOL	Vertical take-off and landing		
W		Y	
WCG	Weight classification group	Z	
WPR	Waypoint reporting		
WTC	Wake turbulence category		
		ZFM	Zero fuel mass